

ON POLYZOMAL CURVES.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLXII. (1872), pp. 147–162. Received January 18,—Read February 8, 1872.]

Part I. (Nos. 1 to 55).—On Polyzomal Curves in General.

1 to 4. Definition and Preliminary Remarks.

Article 1. As already mentioned, U , V , &c. denote rational and integral functions $(* | x, y, z)^r$, all of the same degree r in the coordinates (x, y, z) , and the equation

$$\sqrt{U} + \sqrt{V} + \&c. = 0$$

then belongs to a polyzomal curve; viz., if the number of the zomes $\sqrt{U}$, $\sqrt{V}$, &c. is $= \nu$, then we have a ν -zomal curve. The radicals, or any of them, may contain rational factors, or be of the form $P\sqrt{Q}$; but in speaking of the curve as a ν -zomal, it is assumed that any two terms, such as $P\sqrt{Q} + P'\sqrt{Q}$, involving the same radical $\sqrt{Q}$, are united into a single term, so that the number of distinct radicals is always $= \nu$; in particular (r being even), it is assumed that there is only one rational term P .

But the ordinary case, and that which is almost exclusively attended to, is that in which the radicals $\sqrt{U}$, $\sqrt{V}$, &c. are distinct irreducible radicals without rational factors.

Article 2. The curves $U = 0$, $V = 0$, &c. are said to be the *zomal curves*, or simply the *zomals* of the polyzomal curve $\sqrt{U} + \sqrt{V} + \&c. = 0$; more strictly, the term zomal would be applied to the functions U , V , &c. It is to be noticed, that although the form $\sqrt{U} + \sqrt{V} + \&c. = 0$ is equally general with the form $\sqrt{lU} + \sqrt{mV} + \&c. = 0$ (in fact, in the former case, the functions U , V , &c. are considered as implicitly containing the constant factors l , m , &c., which are expressed in the latter case), yet it is frequently convenient to express these factors, and thus write the equation in the form $\sqrt{lU} + \sqrt{mV} + \&c. = 0$.

For instance, in speaking of any given curves $U = 0$, $V = 0$, &c., we are apt, disregarding the constant factors which they may involve, to consider U , V , &c. as given functions; but in this case the general equation of the polyzomal with the zomals $U = 0$, $V = 0$, &c., is of course $\sqrt{lU} + \sqrt{mV} + \&c. = 0$.

Article 3. Anticipating in regard to the cases $\nu = 1$, $\nu = 2$, the remark which will be presently made in regard to the ν -zomal, that $\sqrt{U} + \sqrt{V} + \&c. = 0$ is the curve represented by the rationalised form of this equation, the monozomal curve $\sqrt{U} = 0$ is merely the curve $U = 0$, viz., this is any curve whatever $U = 0$ of the order r ; and similarly, the bizomal curve $\sqrt{U} + \sqrt{V} = 0$ is merely the curve $U - V = 0$, viz. this is any curve whatever $U - V = 0$, of the order r ; the zomal curves $U = 0$, $V = 0$, taken separately, are not

curves standing in any special relation to the curve in question $= 0$, but $U = 0$ may be any curve whatever of the order r , and then $V = 0$ is a curve of the same order r , in involution with the two curves $U = 0, V = 0$; we may, in fact, write the equation $U - V = 0$ under the bizomal form $\sqrt{U} + \sqrt{V} + \sqrt{U} = 0$. In the case r even, we may, however, notice the bizomal curve $P + \sqrt{U} = 0$ (P a rational function of the degree $\frac{1}{2}r$); the rational equation is here $\Omega = U - P^2 = 0$, that is $U = \Omega + P^2$, viz., P is any curve whatever of the order $\frac{1}{2}r$, and $U = 0$ is a curve of the order r , touching the given curve $\Omega = 0$ at each of its $\frac{1}{2}r^2$ intersections with the curve $P = 0$.

I further remark that the order of the ν -zomal curve $\sqrt{U} + \sqrt{V} + \&c. = 0$ is $= 2^{\nu-1}r$; this is right in the case of the bizomal curve $\sqrt{U} + \sqrt{V} = 0$, the order being $= r$, but it fails for the monozomal curve $\sqrt{U} = 0$, the order being in this case r , instead of $\frac{1}{2}r$, as given by the formula. The two unimportant and somewhat exceptional cases $\nu = 1, \nu = 2$, are thus disposed of, and in all that follows (except in so far as this is in fact applicable to the cases just referred to), ν may be taken to be $= 3$ at least.

Article 4. It is to be throughout understood that by the curve $\sqrt{U} + \sqrt{V} + \&c. = 0$ is meant the curve represented by the rationalised equation

$$\text{Norm}(\sqrt{U} + \sqrt{V} + \&c.) = 0,$$

viz. the Norm is obtained by attributing to all but one of the zomes $\sqrt{U}, \sqrt{V}, \&c.$, each of the two signs $+, -$, and multiplying together the several resulting values of the polyzome; in the case of a ν -zomal curve, the number of factors is thus $= 2^{\nu-1}$ (whence, as each factor is of the degree $\frac{1}{2}r$, the order of the curve is $2^{\nu-1} \cdot \frac{1}{2}r = 2^{\nu-2}r$, as mentioned above). I expressly mention that, as regards the polyzomal curve, we are not in any wise concerned with the signs of the radicals, which signs are and remain essentially indeterminate; the equation $\sqrt{U} + \sqrt{V} + \&c. = 0$, is a mere symbol for the rationalised equation, $\text{Norm}(\sqrt{U} + \sqrt{V} + \&c.) = 0$.

5 to 12. The Branches of a Polyzomal Curve.

Article 5. But we may in a different point of view attend to the signs of the radicals; if for all values of the coordinates we take the symbol $\sqrt{}$, and consider $\sqrt{U}, \sqrt{V}, \&c.$ as signifying determinately, say the positive values of $\sqrt{U}, \sqrt{V}, \&c.$; then each of the several equations $\pm\sqrt{U} \pm \sqrt{V} + \&c. = 0$, or, fixing at pleasure one of the signs, suppose that prefixed to $\sqrt{U}$, then each of the several equations $\sqrt{U} \pm \sqrt{V} \pm \&c. = 0$, will belong to a branch of the polyzomal curve: a ν -zomal curve has thus $2^{\nu-1}$ branches corresponding to the $2^{\nu-1}$ values respectively of the polyzome. The separation of the branches depends on the precise fixation of the significations of $\sqrt{U}, \sqrt{V}, \&c.$, and in regard hereto some further explanation is necessary.

Article 6. When U is real and positive, $\sqrt{U}$ may be taken to be, in the ordinary sense, the positive value of $\sqrt{U}$, and so when U is real and negative, $\sqrt{U}$ may be taken to be $= i$ into the positive value of $\sqrt{-U}$; and the like as regards $\sqrt{V}, \&c.$ The functions $U, V, \&c.$ are assumed to be real functions of the coordinates; hence, for any real values of the coordinates, $U, V, \&c.$ are real positive or negative quantities, and the significations of $\sqrt{U}, \sqrt{V}, \&c.$ are completely determined.

Article 7. But the coordinates may be imaginary. In this case the functions $U, V, \&c.$ will for any given values of the coordinates acquire each of them a determinate, in general

imaginary, value. If for all real values whatever of α, β , we select once for all one of the two opposite values of $\sqrt{\alpha + \beta i}$, calling it the positive value, and representing it by $\sqrt{\alpha + \beta i}$, then, for any particular values of the coordinates, U being $= \alpha + \beta i$, the value of $\sqrt{U}$ may be taken to be $= \sqrt{\alpha + \beta i}$; and the like as regards $\sqrt{V}$, &c. $\sqrt{U}$, $\sqrt{V}$, &c. have thus each of them a determinate signification for any values whatever, real or imaginary, of the coordinates. The coordinates of a given point on the curve $\sqrt{U} + \sqrt{V} + \&c. = 0$, will in general satisfy only one of the equations $\sqrt{U} \pm \sqrt{V} \pm \&c. = 0$; that is, the point will belong to one (but in general only one) of the $2^{\nu-1}$ branches of the curve: the entire series of points the coordinates of which satisfy any one of the $2^{\nu-1}$ equations, will constitute the branch corresponding to that equation.

Article 8. The signification to be attached to the expression $\sqrt{\alpha + \beta i}$ should agree with that previously attached to the like symbol in the case of a positive or negative real quantity; and it should, as far as possible, be subject to the condition of continuity, viz., as $\alpha + \beta i$ passes continuously to $\alpha' + \beta' i$, so $\sqrt{\alpha + \beta i}$ should pass continuously to $\sqrt{\alpha' + \beta' i}$; but (as is known) it is not possible to satisfy universally this condition of continuity; viz., if for facility of explanation we consider (α, β) as the coordinates of a point in a plane, and imagine this point to describe a closed curve surrounding the origin or point $(0, 0)$, then it is not possible so to define $\sqrt{\alpha + \beta i}$ that this quantity, varying continuously as the point moves along the curve, shall, when the point has made a complete circuit, resume its original value. The signification to be attached to $\sqrt{\alpha + \beta i}$ is thus in some measure arbitrary, and it would appear that the division of the curve into branches is affected by a corresponding arbitrariness, but this arbitrariness relates only to the imaginary branches of the curve: the notion of a real branch is perfectly definite.

Article 9. It would seem that a branch may be impossible for any series whatever of points real or imaginary. Thus, in the bizomal curve $\sqrt{U} + \sqrt{V} = 0$, the branch $\sqrt{U} + \sqrt{V} = 0$ if $U - V$ is impossible. In fact, for any point whatever, real or imaginary, of the curve, we have $U = V$, and therefore $\sqrt{U} = \sqrt{V}$; the point thus belongs to the other branch $\sqrt{U} - \sqrt{V} = 0$, not to the branch $\sqrt{U} + \sqrt{V} = 0$; the only points belonging to the last-mentioned branch are the isolated points for which simultaneously $\sqrt{U} = 0$, $\sqrt{V} = 0$; viz., the points of intersection of the two curves $U = 0$, $V = 0$.

Article 10. It is not clear to me whether the case is the same in regard to the branch $\sqrt{U} + \sqrt{V} + \sqrt{W} = 0$ of a trizomal curve. In fact, for each point of the curve $\sqrt{U} + \sqrt{V} + \sqrt{W} = 0$ we have $(U - V - W)^2 = 4VW$, and therefore, $U - V - W = \pm 2\sqrt{V}\sqrt{W}$: there may very well be points for which the sign is $+$; that is, points for which $U = V + W + 2\sqrt{VW}$, and for these points we have $\pm\sqrt{U} = \sqrt{V} + \sqrt{W}$; for real values of the coordinates the sign on the left hand must be $+$ (for otherwise the two sides would have opposite signs), but there is no apparent reason, or at least no obviously apparent reason, why this should be so for imaginary values of the coordinates, and if the sign be in fact $-$ then the point will belong to the branch $-\sqrt{U} + \sqrt{V} + \sqrt{W} = 0$.

Article 11. But the branch in question is clearly impossible for any series of real points; so that, leaving it an open question whether the epithet "impossible" is to be understood to mean impossible for any series of real points (that is, as a mere synonym of imaginary), or whether it is to mean impossible for any series of points, real or imaginary, whatever, I say that in a ν -zomal curve some of the branches are or may be impossible, and that there is at least one impossible branch, viz., the branch $\sqrt{U} + \sqrt{V} + \&c. = 0$.

Article 12. For the purpose of referring to any branch of a polyzomal curve it will be convenient to consider $\sqrt{U}$ as signifying determinately $+\sqrt{U}$, or else $-\sqrt{U}$; and the like as regards $\sqrt{V}$, &c., but without any identity or relation between the signs prefixed to the $\sqrt{U}$, $\sqrt{V}$, &c., respectively; the equation $\sqrt{U} + \sqrt{V} + \&c. = 0$, so understood, will be that of any determinate branch of the ν -zomal curve.

13 to 17. The Points common to Two Branches of a Polyzomal Curve.

Article 13. I say that two branches of a ν -zomal curve have in common $2^{\nu-2}r^2$ points, and that these are points belonging to a $(\nu - 1)$ -zomal curve.

Article 14. In fact, the equations of the two branches may be taken to be

$$\sqrt{U} + \sqrt{V} + \sqrt{W} + \&c. = 0, \quad \sqrt{U} - \sqrt{V} + \sqrt{W} + \&c. = 0,$$

(viz. the two branches differ only as to the sign of the radical $\sqrt{V}$); the two equations give $\sqrt{V} = 0$, that is $V = 0$, and

$$\sqrt{U} + \sqrt{W} + \&c. = 0,$$

which last equation is that of a $(\nu - 1)$ -zomal curve. The intersections of the two branches are thus the $2^{\nu-2}r \cdot r = 2^{\nu-2}r^2$ intersections of the curve $V = 0$ with the $(\nu - 1)$ -zomal curve $\sqrt{U} + \sqrt{W} + \&c. = 0$.

Article 15. If the two branches agree as to the sign of $\sqrt{U}$, but differ as to the sign of $\sqrt{V}$, then the points common to the two branches are points belonging to the $(\nu - 1)$ -zomal curve $\sqrt{U} + \sqrt{W} + \&c. = 0$; and the like if the branches differ as to the sign of any other radical $\sqrt{V}$. So, if the branches differ as to the signs of any two radicals $\sqrt{V}$, $\sqrt{W}$, the points common to the two branches are points belonging to the $(\nu - 2)$ -zomal curve $\sqrt{U} + \&c. = 0$; and so on. But the statement as originally made, that the points common to two branches are points belonging to a $(\nu - 1)$ -zomal, assumes that the two branches differ as to the sign of a single radical only.

Article 16. The several branches meet by pairs at $2^{\nu-3}r^2$ points on each of the curves $U = 0$, $V = 0$, $W = 0$, &c. respectively; and the like as to the several curves which are represented by the equations obtained by attributing to the radicals $\sqrt{U}$, $\sqrt{V}$, &c. each of the two signs $+$ and $-$, and equating to zero the sum of any even number of the zomes; for instance, at the intersections of the curves $U = 0$, $V = 0$, we have $\sqrt{U} = 0$, $\sqrt{V} = 0$, and therefore $\sqrt{U} + \sqrt{V} + \sqrt{W} + \&c. = 0$, $\sqrt{U} + \sqrt{V} - \sqrt{W} + \&c. = 0$, &c.; that is, the several branches which contain the term $+\sqrt{U} + \sqrt{V}$ all pass through the r^2 intersections of the curves $U = 0$, $V = 0$; and so the several branches which contain the term $+\sqrt{U} - \sqrt{V}$ all pass through the same r^2 intersections: the number of branches which contain $+\sqrt{U} + \sqrt{V}$ is $2^{\nu-3}$, and the number of branches which contain $+\sqrt{U} - \sqrt{V}$ is $2^{\nu-3}$, so that the r^2 intersections of $U = 0$, $V = 0$ are each of them a point on $2^{\nu-3} + 2^{\nu-3} = 2^{\nu-2}$ branches; that is, on a pair of branches. And similarly at the r^2 intersections of any other pair of zomal curves $U = 0$, $W = 0$, $V = 0$, $W = 0$, &c. respectively. Also at the r^2 intersections of $U = 0$ with the curve $\sqrt{V} + \sqrt{W} + \&c. = 0$, the several branches which contain $+\sqrt{U}$ all meet the several branches which contain $-\sqrt{U}$; the number of each set of branches being $2^{\nu-2}$, we have thus at each of the $2^{\nu-2}r^2$ intersections in question a meeting of a pair of branches.

Article 17. The points common to two branches of a ν -zomal are singular points of the rationalised curve $\text{Norm}(\sqrt{U} + \sqrt{V} + \&c.) = 0$; but I do not at present inquire as to the nature of the singularity at any such point.

18 to 21. Singularities of a ν -zomal Curve.

Article 18. Consider a branch $\sqrt{U} + \sqrt{V} + \sqrt{W} + \&c. = 0$, and any point whatever on this branch; we have at this point $\sqrt{U} = -(\sqrt{V} + \sqrt{W} + \&c.)$, that is

$$U = (\sqrt{V} + \sqrt{W} + \&c.)^2,$$

and therefore

$$U - V - W - \&c. = 2(\sqrt{VW} + \&c.),$$

where the term on the right hand includes the products of the radicals $\sqrt{V}$, $\sqrt{W}$, $\&c.$ taken two and two together. Hence, squaring and transposing,

$$(U - V - W - \&c.)^2 - 4(VW + \&c.) = 8(\sqrt{VWX} + \&c.),$$

where the term on the right hand now includes the products of the radicals taken three and three together. Proceeding in this manner we ultimately obtain an equation wherein the right hand side contains only a single radical, say $\sqrt{\Theta}$, the left hand side being a rational function of (x, y, z) ; squaring, we obtain the rationalised equation $F = 0$, where F is a rational function of (x, y, z) , of the order $2^{\nu-1}r$.

Article 19. The rationalised equation may be obtained in a different form as follows: write the equation in the form $\sqrt{U} + \sqrt{V} = -(\sqrt{W} + \&c.)$; squaring, we obtain

$$U + V + 2\sqrt{UV} = (\sqrt{W} + \&c.)^2,$$

and thence $2\sqrt{UV} = -(U + V) + (\sqrt{W} + \&c.)^2$, an equation containing on the right hand side a single radical less by one than the original equation; and so we ultimately obtain an equation containing on one side a single radical only, which on being squared gives the rationalised equation.

Article 20. In the case of a trizomal curve $\sqrt{U} + \sqrt{V} + \sqrt{W} = 0$, the rationalised equation is

$$U^2 + V^2 + W^2 - 2UV - 2UW - 2VW = 0,$$

that is

$$(U + V + W)^2 - 4(VW + WU + UV) = 0,$$

or, as this may also be written,

$$(\sqrt{U} + \sqrt{V} + \sqrt{W})(\sqrt{U} + \sqrt{V} - \sqrt{W})(\sqrt{U} - \sqrt{V} + \sqrt{W})(-\sqrt{U} + \sqrt{V} + \sqrt{W}) = 0.$$

In the case of a tetrazomal curve $\sqrt{U} + \sqrt{V} + \sqrt{W} + \sqrt{X} = 0$, the rationalised equation is of the order $8r$, and the number of terms is very large.

Article 21. If in the rationalised equation of the ν -zomal we consider any point whatever of the curve $\sqrt{U} + \sqrt{V} + \&c. = 0$, we have of course $F = 0$; and moreover $dF = 0$, where dF is the differential of F taken in regard to the coordinates (x, y, z) in any manner whatever. But this does not establish that the point in question is a singular point of the rationalised curve; for in order that it may be a singular point, we require the equation $dF = 0$ for all values whatever of $dx : dy : dz$, and this is not in general the case. In fact, for a point on the branch $\sqrt{U} + \sqrt{V} + \&c. = 0$, the equations $F = 0$, $dF = 0$ imply only that this point belongs to the branch in question, and not that it is a singular point thereof.

22 to 27. Special Case where all the Zomals have a Common Point or Points.

Article 22. Suppose that the zomals $U = 0$, $V = 0$, $W = 0$, &c. all pass through the same point or points; this will be the case if U , V , W , &c. all contain as a factor one and the same function Θ of the coordinates, say if $U = \Theta \cdot U'$, $V = \Theta \cdot V'$, $W = \Theta \cdot W'$, &c., where Θ is a function of the degree m , and U' , V' , W' , &c. are functions of the degree $r - m$. We have then

$$\sqrt{U} + \sqrt{V} + \&c. = \sqrt{\Theta}(\sqrt{U'} + \sqrt{V'} + \&c.).$$

The rationalised equation is thus $\Theta^{2^{\nu-2}} \cdot \text{Norm}(\sqrt{U'} + \sqrt{V'} + \&c.) = 0$; that is, the rationalised equation contains the factor $\Theta^{2^{\nu-2}} = 0$, or the curve $\text{Norm}(\sqrt{U} + \sqrt{V} + \&c.) = 0$ contains the curve $\Theta = 0$ as an $2^{\nu-2}$ -tuple curve; and omitting this factor, the remaining factor $\text{Norm}(\sqrt{U'} + \sqrt{V'} + \&c.) = 0$ is a curve of the order $2^{\nu-2}(r - m)$; that is, the order of the curve is depressed from $2^{\nu-2}r$ to $2^{\nu-2}(r - m)$, viz. the depression is $= 2^{\nu-2}m$.

Article 23. If the zomals have a common point of the multiplicity m , then the several branches of the ν -zomal curve all pass through this point; and the tangents at the common point are given by the equation obtained from the equation of any branch by substituting therein for the coordinates their values in terms of a parameter, and equating to zero the terms of the lowest degree in the parameter. For instance, if the common point be the point $(x = 0, y = 0)$, then writing $x = x_1 t$, $y = y_1 t$, where (x_1, y_1) are the coordinates of a point on the line joining the origin to the point in question, and t is a parameter, we obtain from the equation $\sqrt{U} + \sqrt{V} + \&c. = 0$ an equation wherein each of the functions U , V , &c. contains the factor t^m , and wherefore $\sqrt{U}$, $\sqrt{V}$, &c. each contain the factor $t^{\frac{1}{2}m}$; the equation thus becomes divisible by $t^{\frac{1}{2}m}$, and equating to zero the quotient, we obtain the equation of the tangents at the common point.

Article 24. If the zomals have two common points, each of the multiplicity m , then the rationalised equation contains the factor $\Theta^{2^{\nu-2}}$, where Θ is now the product of the two linear functions which represent the two common points, viz. Θ is of the degree $2m$; the depression of order is thus $= 2 \cdot 2^{\nu-2}m = 2^{\nu-1}m$. And similarly if the zomals have k common points, each of the multiplicity m , the depression of order is $= k \cdot 2^{\nu-2}m$.

Article 25. If the zomals have a common curve of the order m , then Θ is of the degree m , and the depression of order is $= 2^{\nu-2}m$, as in Art. 22. But if the common curve is a multiple curve on each of the zomals, say a k -tuple curve, then the depression is $= k \cdot 2^{\nu-2}m$.

Article 26. The case where the zomals have a common point or points is a particular case of a more general theory, which will be presently considered; but it is convenient to notice it here, as it affords a simple explanation of the depression of order which takes place in certain cases.

Article 27. In particular, if the zomals $U = 0$, $V = 0$, $W = 0$, &c. all pass through the same point, and this point is a simple point on each of the zomals, then the several branches of the ν -zomal curve all pass through this point, and the tangents at the point are given by the equation $\sqrt{L} + \sqrt{M} + \sqrt{N} + \&c. = 0$, where $L = 0$, $M = 0$, $N = 0$, &c. are the tangents of the zomals at the common point; that is, the tangents at the common point form a ν -zomal curve of the first order, the zomals being the tangents $L = 0$, $M = 0$, $N = 0$, &c. respectively.

28 to 37. Depression of Order of the ν -zomal Curve from the Ideal Factor $z = 0$.

Article 28. I proceed to the theory of the depression of order from the ideal factor $z = 0$. The coordinates are throughout (x, y, z) ; and when the equation is given in terms of the Cartesian coordinates (x, y) , it is to be understood that the coordinates are made homogeneous by the introduction of z in the proper power; thus the equation of a circle is written $(x - \alpha z)^2 + (y - \beta z)^2 = \gamma^2 z^2$, and the line infinity is $z = 0$. A curve ideally contains the line infinity when the equation in its homogeneous form contains the factor $z = 0$.

Article 29. Consider the trizomal curve $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$, where U, V, W are functions of the degree r , and where for greater generality the constant factors l, m, n are expressed. Suppose that the three zomals $U = 0, V = 0, W = 0$ all touch the line infinity; viz. that we have $U = (Az + \dots)^2, V = (Bz + \dots)^2, W = (Cz + \dots)^2$, where the omitted terms are of the degree r in (x, y, z) , but do not contain any terms of the form $(Az)^2, (Bz)^2, (Cz)^2$ respectively; that is, the terms of the degree r in (x, y) form perfect squares. We have thus $\sqrt{U} = Az + \dots, \sqrt{V} = Bz + \dots, \sqrt{W} = Cz + \dots$, and the equation of the branch becomes

$$l(Az + \dots) + m(Bz + \dots) + n(Cz + \dots) = 0,$$

that is

$$(lA + mB + nC)z + \&c. = 0.$$

Article 30. The line infinity will thus be a factor of the branch if $lA + mB + nC = 0$; and if this condition be satisfied, then the branch ideally contains the line infinity. The rationalised equation will thus contain the factor $z = 0$, corresponding to each branch which ideally contains the line infinity; and the depression of order is equal to the number of such branches.

Article 31. In the case of the trizomal curve, if the condition $lA + mB + nC = 0$ is satisfied, then each of the four branches ideally contains the line infinity; the rationalised equation thus contains the factor z^4 , and the depression of order is $= 4$; the order of the curve is thus reduced from $4r$ to $4r - 4$.

Article 32. But if the condition $lA + mB + nC = 0$ is not satisfied, then the line infinity is not a factor of any branch; and the order of the curve remains $= 4r$. And similarly for the general ν -zomal curve: if the zomals all touch the line infinity, and if the coefficients $l, m, \&c.$ satisfy a certain condition, then each of the $2^{\nu-1}$ branches ideally contains the line infinity, and the depression of order is $= 2^{\nu-1}$; but if this condition is not satisfied, there is no depression of order.

Article 33. The condition $lA + mB + nC = 0$ may be geometrically interpreted as follows: $A = 0, B = 0, C = 0$ are the tangents at infinity to the zomals $U = 0, V = 0, W = 0$ respectively; the condition $lA + mB + nC = 0$ expresses that these three tangents meet in a point; that is, that the three points of contact of the zomals with the line infinity lie on a right line. But if the zomals are circles, then A, B, C are constant, and the condition reduces to $lA + mB + nC = 0$, which is a relation between the coefficients l, m, n .

Article 34. In the case where the zomals are circles, the condition $lA + mB + nC = 0$ is satisfied if l, m, n are proportional to the squares of the radii of the circles; for then $l = kR^2, m = kR'^2, n = kR''^2$, where R, R', R'' are the radii; and $A = 1/R, B = 1/R', C = 1/R''$; so that $lA + mB + nC = k(R + R' + R'')$, which is not in general zero. But if the circles have

their centres on a right line, and if l, m, n are properly determined, the condition may be satisfied.

Article 35. The case where the zomals all touch the line infinity is a particular case of the more general case where the zomals all pass through a common point on the line infinity; but the theory is somewhat different, as the depression of order depends not only on the existence of the common point, but also on the relation between the coefficients l, m, n .

Article 36. If the zomals all pass through two common points on the line infinity, then each zomal contains the line infinity as a factor, and the depression of order is greater than in the case of a single common point. Thus, if each zomal contains the line infinity as a simple factor, we have $U = z \cdot U', V = z \cdot V', W = z \cdot W'$, where U', V', W' are of the degree $r - 1$; and the equation of the branch is $\sqrt{z}(\sqrt{lU'} + \sqrt{mV'} + \sqrt{nW'}) = 0$, which gives $z = 0$ as a factor of the rationalised equation; the depression of order is thus $= 2^{\nu-1}$, as before.

Article 37. But if each zomal contains the line infinity as a double factor, we have $U = z^2 U', V = z^2 V', W = z^2 W'$, where U', V', W' are of the degree $r - 2$; and the equation of the branch is $z(\sqrt{lU'} + \sqrt{mV'} + \sqrt{nW'}) = 0$, which gives $z^{2^{\nu-1}} = 0$ as a factor of the rationalised equation; the depression of order is thus $= 2 \cdot 2^{\nu-1} = 2^\nu$.

38 and 39. On the Trizomal Curve and the Tetrzomal Curve.

Article 38. The trizomal curve

$$\sqrt{U} + \sqrt{V} + \sqrt{W} = 0$$

has for its rationalised form of equation

$$U^2 + V^2 + W^2 - 2VW - 2WU - 2UV = 0;$$

or as this may also be written,

$$(1, 1, 1, -1, -1, -1 \mid \sqrt{U}, \sqrt{V}, \sqrt{W})^2 = 0;$$

and we may from this rational equation verify the general results applicable to the case in hand, viz., that the trizomal is a curve of the order $2r$, and that

$$U = 0, \text{ at each of its } r^2 \text{ intersections with } V - W = 0,$$

$$V = 0, \text{ at its } r^2 \text{ intersections with } W - U = 0,$$

$$W = 0, \text{ at its } r^2 \text{ intersections with } U - V = 0,$$

respectively touch the trizomal. There are not, in general, any nodes or cusps, and the order being $= 2r$, the class is $= 2r(2r - 1)$.

Article 39. The tetrzomal curve

$$\sqrt{U} + \sqrt{V} + \sqrt{W} + \sqrt{T} = 0$$

has for its rationalised form of equation

$$(U^2 + V^2 + W^2 + T^2 - 2UV - 2UW - 2UT - 2VW - 2VT - 2WT)^2 - 64UVWT = 0,$$

and we may hereby verify the fundamental properties, viz., that the tetrazomal is a curve of the order $4r$, touched by each of the zomals $U = 0$, $V = 0$, $W = 0$, $T = 0$ in $2r^2$ points, viz., by $U = 0$ at its intersections with $\sqrt{V} + \sqrt{W} + \sqrt{T} = 0$, that is, $V^2 + W^2 + T^2 - 2VW - 2VT - 2WT = 0$; (and the like as regards the other zomals), and having $3r^4$ nodes, viz., these are the intersections of $(\sqrt{U} + \sqrt{V} = 0, \sqrt{W} + \sqrt{T} = 0)$, $(\sqrt{U} + \sqrt{W} = 0, \sqrt{V} + \sqrt{T} = 0)$, $(\sqrt{U} + \sqrt{T} = 0, \sqrt{V} + \sqrt{W} = 0)$, or, what is the same thing, the intersections of $(U - V = 0, W - T = 0)$, $(U - W = 0, V - T = 0)$, $(U - T = 0, V - W = 0)$. There are not in general any cusps, and the class is thus $= 4r(4r - 1) - 6r^4$, $= 16r^2 - 4r - 6r^4$.

40 and 41. On the Intersection of two ν -Zomals having the same Zomal Curves.

Article 40. Without going into any detail, I may notice the question of the intersection of two ν -zomals which have the same zomal curves: say the two trizomals $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$, $\sqrt{l'U} + \sqrt{m'V} + \sqrt{n'W} = 0$, or two similarly related tetrazomals. For the trizomals, writing the equations under the form

$$\sqrt{U} + \sqrt{V} + \sqrt{W} = 0, \quad \sqrt{l}\sqrt{U} + \sqrt{m'}\sqrt{V} + \sqrt{n'}\sqrt{W} = 0,$$

then, when these equations are considered as existing simultaneously, we may, without loss of generality, attribute to the radicals $\sqrt{U}$, $\sqrt{V}$, $\sqrt{W}$, the same values in the two equations respectively; but doing so, we must in the second equation successively attribute to all but one of the radicals $\sqrt{l'}$, $\sqrt{m'}$, $\sqrt{n'}$, each of its two opposite values. For the intersections of the two curves we have thus

$$\sqrt{U} : \sqrt{V} : \sqrt{W} = \sqrt{m'} - \sqrt{n'} : \sqrt{n'} - \sqrt{l'} : \sqrt{l'} - \sqrt{m'},$$

viz., this is one of a system of four equations, obtained from it by changes of sign, say in the radicals $\sqrt{m'}$ and $\sqrt{n'}$. Each of the four equations gives a set of r^2 points; we have thus the complete number, $= 4r^2$, of the points of intersection of the two curves.

Article 41. But take, in like manner, two tetrazomal curves; writing their equations in the form

$$\begin{aligned} \sqrt{U} + \sqrt{V} + \sqrt{W} + \sqrt{T} &= 0, \\ \sqrt{l'}\sqrt{U} + \sqrt{m'}\sqrt{V} + \sqrt{n'}\sqrt{W} + \sqrt{p'}\sqrt{T} &= 0, \end{aligned}$$

then $\sqrt{U}$, $\sqrt{V}$, $\sqrt{W}$, $\sqrt{T}$ may be considered as having the same values in the two equations respectively, but we must in the second equation attribute successively, say to $\sqrt{m'}$, $\sqrt{n'}$, $\sqrt{p'}$, each of their two opposite values. For the intersections of the two curves we have

$$\begin{aligned} (\sqrt{m'} - \sqrt{l'})\sqrt{V} + (\sqrt{n'} - \sqrt{l'})\sqrt{W} + (\sqrt{p'} - \sqrt{l'})\sqrt{T} &= 0, \\ (\sqrt{l'} - \sqrt{m'})\sqrt{U} + (\sqrt{n'} - \sqrt{m'})\sqrt{W} + (\sqrt{p'} - \sqrt{m'})\sqrt{T} &= 0, \end{aligned}$$

viz., this is one of a system of eight similar pairs of equations, obtained therefrom by changes of sign of the radicals $\sqrt{m'}$, $\sqrt{n'}$, $\sqrt{p'}$. The equations represent each of them a trizomal curve, of the order $2r$; the two curves intersect therefore in $4r^2$ points, and if each of these was a point of intersection of the two tetrazomals, we should have in all $8 \times 4r^2 = 32r^2$ intersections. But the tetrazomals are each of them a curve of the order $4r$, and they intersect therefore in only $16r^2$ points. The explanation is, that not all the $4r^2$ points, but only $2r^2$ of them are

intersections of the tetrazomals. In fact, to find all the intersections of the two trizomals, it is necessary in their two equations to attribute opposite signs to one of the radicals $\sqrt{U}$, $\sqrt{V}$; we obtain $2r^2$ intersections from the equations as they stand, the remaining $2r^2$ intersections from the two equations after we have in the second equation reversed the sign, say of $\sqrt{V}$.

Now, from the two equations as they stand we can pass back to the two tetrazomal equations, and the first-mentioned $2r^2$ points are thus points of intersection of the two tetrazomal curves: from the two equations after such reversal of the sign of $\sqrt{V}$, we cannot pass back to the two tetrazomal equations, and the last-mentioned $2r^2$ points are thus not points of intersection of the two tetrazomal curves. The number of intersections of the two curves is thus $8 \times 2r^2 = 16r^2$, as it should be.

42 to 45. The Theorem of the Decomposition of a Tetrazomal Curve.

Article 42. I consider the tetrazomal curve

$$\sqrt{lU} + \sqrt{mV} + \sqrt{nW} + \sqrt{pT} = 0,$$

where the zomal curves are in involution, that is, where we have an identical relation,

$$aU + bV + cW + dT = 0;$$

and I proceed to show that if l, m, n, p satisfy the relation

$$\frac{l}{a} + \frac{m}{b} + \frac{n}{c} + \frac{p}{d} = 0,$$

the curve breaks up into two trizomals. In fact, writing the equation under the form

$$(\sqrt{l}\sqrt{U} + \sqrt{m}\sqrt{V} + \sqrt{n}\sqrt{W})^2 - pT = 0,$$

and substituting for T its value, in terms of U, V, W , this is

$$(ld + pa)U + (md + pb)V + (nd + pc)W + 2\sqrt{mn}d\sqrt{VW} + 2\sqrt{nl}d\sqrt{WU} + 2\sqrt{lm}d\sqrt{UV} = 0;$$

or, considering the left-hand side as a quadric function of $(\sqrt{U}, \sqrt{V}, \sqrt{W})$, the condition for its breaking up into factors is

$$\begin{vmatrix} ld + pa & d\sqrt{mn} & d\sqrt{nl} \\ d\sqrt{mn} & md + pb & d\sqrt{lm} \\ d\sqrt{nl} & d\sqrt{lm} & nd + pc \end{vmatrix} = 0,$$

that is

$$d^2(lbcd + mcda + ndab + pabc) = 0,$$

or finally, the condition is

$$\frac{l}{a} + \frac{m}{b} + \frac{n}{c} + \frac{p}{d} = 0.$$

Article 43. Multiplying by $\frac{1}{d}(ld + pa)$, and observing that in virtue of the relation we have

$$\frac{1}{d}(ld + pa) \cdot \frac{1}{d}(md + pb) = \frac{1}{d^2}(lm + pa \cdot \frac{m}{b} \cdot \frac{b}{a} + \dots),$$

the equation becomes

$$\left(\sqrt{ld+pa}\sqrt{U} + \sqrt{md+pb}\sqrt{V} + \sqrt{nd+pc}\sqrt{W}\right)^2 = \left(\sqrt{\frac{p}{d}}\sqrt{bV-cW}\right)^2,$$

or as this is more conveniently written

$$\left(\sqrt{l_1}\sqrt{U} + \sqrt{m_1}\sqrt{V} + \sqrt{n_1}\sqrt{W}\right)\left(\sqrt{l_2}\sqrt{U} + \sqrt{m_2}\sqrt{V} + \sqrt{n_2}\sqrt{W}\right) = 0,$$

an equation breaking up into two equations, which may be represented by

$$\sqrt{l_1U} + \sqrt{m_1V} + \sqrt{n_1W} = 0, \quad \sqrt{l_2U} + \sqrt{m_2V} + \sqrt{n_2W} = 0,$$

where

$$\begin{aligned} \sqrt{l_1} &= \sqrt{ld+pa} - \sqrt{\frac{p}{d}}\sqrt{bc}, & \sqrt{l_2} &= \sqrt{ld+pa} + \sqrt{\frac{p}{d}}\sqrt{bc}, \\ \sqrt{m_1} &= \sqrt{md+pb} - \sqrt{\frac{p}{d}}\sqrt{ca}, & \sqrt{m_2} &= \sqrt{md+pb} + \sqrt{\frac{p}{d}}\sqrt{ca}, \\ \sqrt{n_1} &= \sqrt{nd+pc} - \sqrt{\frac{p}{d}}\sqrt{ab}, & \sqrt{n_2} &= \sqrt{nd+pc} + \sqrt{\frac{p}{d}}\sqrt{ab}, \end{aligned}$$

where, in the expressions for $\sqrt{l_1}$, &c., the signs of the radicals may be taken determinately in any way whatever at pleasure: the only effect of an alteration of sign would in some cases be to interchange the values of $(\sqrt{l_1}, \sqrt{m_1}, \sqrt{n_1})$ with those of $(\sqrt{l_2}, \sqrt{m_2}, \sqrt{n_2})$. The tetrazomal curve thus breaks up into two trizomals.

Article 44. It is to be noticed that we have

$$\frac{l_1}{a} + \frac{m_1}{b} + \frac{n_1}{c} = \frac{1}{d}(ld+pa+md+pb+nd+pc) - \sqrt{\frac{p}{d}}\left(\sqrt{\frac{bc}{a}} + \sqrt{\frac{ca}{b}} + \sqrt{\frac{ab}{c}}\right),$$

that is

$$\frac{l_1}{a} + \frac{m_1}{b} + \frac{n_1}{c} = -\frac{p}{d}\left(\frac{d}{a} + \frac{d}{b} + \frac{d}{c} + 1\right) = 0,$$

and that similarly we have

$$\frac{l_2}{a} + \frac{m_2}{b} + \frac{n_2}{c} = 0.$$

The meaning is, that, taking the trizomal curve $\sqrt{l_1U} + \sqrt{m_1V} + \sqrt{n_1W} = 0$, this regarded as a tetrazomal curve, $\sqrt{l_1U} + \sqrt{m_1V} + \sqrt{n_1W} + \sqrt{0 \cdot T} = 0$, satisfies the condition $\frac{l_1}{a} + \frac{m_1}{b} + \frac{n_1}{c} + \frac{0}{d} = 0$; and the like as to the trizomal curve $\sqrt{l_2U} + \sqrt{m_2V} + \sqrt{n_2W} = 0$.

Article 45. The equation by which the decomposition was effected is, it is clear, one of twelve equivalent equations: four of these are

$$\begin{aligned} \left(\sqrt{ld+pa}\sqrt{U} - \sqrt{\frac{p}{d}}\sqrt{bV-cW}\right)\left(\sqrt{ld+pa}\sqrt{U} + \sqrt{\frac{p}{d}}\sqrt{bV-cW}\right) &= 0, \\ \left(\sqrt{md+pb}\sqrt{V} - \sqrt{\frac{p}{d}}\sqrt{cW-aU}\right)\left(\sqrt{md+pb}\sqrt{V} + \sqrt{\frac{p}{d}}\sqrt{cW-aU}\right) &= 0, \\ \left(\sqrt{nd+pc}\sqrt{W} - \sqrt{\frac{p}{d}}\sqrt{aU-bV}\right)\left(\sqrt{nd+pc}\sqrt{W} + \sqrt{\frac{p}{d}}\sqrt{aU-bV}\right) &= 0, \end{aligned}$$

and the others may be deduced from these by a cyclical permutation of (U, V, W) , (a, b, c) , (l, m, n) , leaving T , d , p unaltered.

46 to 51. Application to the Trizomal; the Theorem of the Variable Zomal.

Article 46. I take the last equation written under the form

$$\left(\sqrt{\frac{m}{b}}\sqrt{U} - \sqrt{\frac{l}{a}}\sqrt{V}\right)^2 = \left(\sqrt{\frac{p}{d}}\sqrt{\frac{bc}{a}}\sqrt{U} + \sqrt{\frac{p}{d}}\sqrt{\frac{ca}{b}}\sqrt{V} + \sqrt{n}\sqrt{W}\right)^2,$$

which, putting therein $p = 0$, is

$$\left(\sqrt{\frac{m}{b}}\sqrt{U} - \sqrt{\frac{l}{a}}\sqrt{V}\right)^2 = nW,$$

which is in fact the trizomal curve,

$$\sqrt{\frac{m}{b}}\sqrt{U} + \sqrt{\frac{l}{a}}\sqrt{V} + \sqrt{n}\sqrt{W} = 0,$$

viz., the trizomal curve $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$, if a, b, c be any quantities connected by the equation

$$\frac{l}{a} + \frac{m}{b} + \frac{n}{c} = 0$$

(the ratios $a : b : c$ thus involving a single arbitrary parameter); and if we take T a function such that $aU + bV + cW + dT = 0$; that is, $T = 0$, any one of the series of curves $aU + bV + cW = 0$, in involution with the given curves $U = 0, V = 0, W = 0$, has its equation expressible in the form

$$\sqrt{\frac{m}{b}}\sqrt{U} + \sqrt{\frac{l}{a}}\sqrt{V} + \sqrt{n}\sqrt{T} = 0;$$

that is, we have the curve (the equation whereof contains a variable parameter) as a zomal of the given trizomal curve $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$; and we have thus from the theorem of the decomposition of a tetrazomal deduced the theorem of the variable zomal of a trizomal. The analytical investigation is somewhat simplified by assuming $p = 0$ ab initio, and it may be as well to repeat it in this form.

Article 47. Starting, then, with the trizomal curve

$$\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0,$$

and writing

$$aU + bV + cW + dT = 0$$

as the definition of T , the coefficients being connected by

$$\frac{l}{a} + \frac{m}{b} + \frac{n}{c} = 0,$$

the equation gives

$$lU + mV + 2\sqrt{lmUV} - nW = 0;$$

or substituting in this equation for W its value in terms of U, V, T , we have

$$(an + cl)U + (bn + cm)V + 2c\sqrt{lmUV} + dnT = 0,$$

which by the given relation between a, b, c , is converted into

$$\sqrt{mU} - \sqrt{lV} + 2c\sqrt{lmUV} + dnT = 0;$$

that is

$$c^2lm \cdot U + l \cdot V - 2ab\sqrt{lmUV} = nT,$$

viz., this is

$$\left(c\sqrt{m}\sqrt{U} - \sqrt{l}\sqrt{V}\right)^2 = nT,$$

or finally

$$\sqrt{m}U + \sqrt{l}V + \sqrt{n}T = 0.$$

Article 48. The result just obtained of course implies that when as above

$$aU + bV + cW + dT = 0, \quad \frac{l}{a} + \frac{m}{b} + \frac{n}{c} = 0,$$

the trizomal curve $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$ can be expressed by means of any three of the four zomals U, V, W, T ; and we may at once write down the four forms

$$\begin{aligned} \sqrt{\frac{m}{b}}\sqrt{U} + \sqrt{\frac{l}{a}}\sqrt{V} + \sqrt{n}\sqrt{W} &= 0, \\ \sqrt{\frac{m}{b}}\sqrt{U} + \sqrt{\frac{l}{a}}\sqrt{V} + \sqrt{n}\sqrt{T} &= 0, \\ \sqrt{\frac{n}{c}}\sqrt{U} + \sqrt{\frac{l}{a}}\sqrt{W} + \sqrt{m}\sqrt{T} &= 0, \\ \sqrt{\frac{n}{c}}\sqrt{V} + \sqrt{\frac{m}{b}}\sqrt{W} + \sqrt{l}\sqrt{T} &= 0, \end{aligned}$$

the last of which is the original equation $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$. It may be added that if the first equation be represented by $\sqrt{l_1U} + \sqrt{m_1V} + \sqrt{p_1T} = 0$, that is, if we have

$$\sqrt{l_1} = \sqrt{\frac{m}{b}}, \quad \sqrt{m_1} = \sqrt{\frac{l}{a}}, \quad \sqrt{p_1} = \sqrt{n},$$

and therefore

$$\frac{l_1}{b} + \frac{m_1}{a} + \frac{p_1}{d} = \frac{1}{bc} \left(\frac{l}{a} + \frac{m}{b} + \frac{n}{c} \right) = 0,$$

or if the second equation be represented by $\sqrt{l_2U} + \sqrt{n_2W} + \sqrt{p_2T} = 0$, that is, if we have

$$\sqrt{l_2} = \sqrt{\frac{m}{b}}, \quad \sqrt{n_2} = -\sqrt{\frac{l}{c}}, \quad \sqrt{p_2} = \sqrt{n},$$

and therefore

$$\frac{l_2}{b} + \frac{n_2}{c} + \frac{p_2}{d} = 0;$$

or if the third equation be represented by $\sqrt{l_3V} + \sqrt{m_3W} + \sqrt{p_3T} = 0$, that is, if we have

$$\sqrt{l_3} = \sqrt{\frac{n}{c}}, \quad \sqrt{m_3} = -\sqrt{\frac{l}{a}}, \quad \sqrt{p_3} = \sqrt{m},$$

and therefore

$$\frac{l_3}{c} + \frac{m_3}{a} + \frac{p_3}{d} = 0;$$

then the equation of the trizomal may also be expressed in the forms

$$\begin{aligned} (\sqrt{l_1U}, \sqrt{m_1V}, \sqrt{p_1T} \mid \sqrt{lU}, \sqrt{mV}, \sqrt{nW}, \sqrt{lT}) &= 0, \\ (\sqrt{l_2U}, \sqrt{n_2W}, \sqrt{p_2T} \mid \sqrt{lU}, \sqrt{mV}, \sqrt{nW}, \sqrt{mT}) &= 0, \\ (\sqrt{l_3V}, \sqrt{m_3W}, \sqrt{p_3T} \mid \sqrt{lU}, \sqrt{mV}, \sqrt{nW}, \sqrt{nT}) &= 0. \end{aligned}$$

Article 49. These equations may, however, be expressed in a much more elegant form. Write

$$a = \frac{a'}{(\beta\gamma\delta)}, \quad b = \frac{b'}{(\gamma\delta\alpha)}, \quad c = \frac{c'}{(\delta\alpha\beta)}, \quad d = \frac{d'}{(\alpha\beta\gamma)},$$

where, for shortness, $(\beta\gamma\delta) = (\beta - \gamma)(\gamma - \delta)(\delta - \beta)$, &c.; (α, β, γ) being arbitrary quantities; or, what is the same thing,

$$a : b : c : d = a'(\beta\gamma\delta) : -b'(\gamma\delta\alpha) : c'(\delta\alpha\beta) : -d'(\alpha\beta\gamma).$$

And if

$$l : m : n = pa'(\beta - \gamma)^2 : qb'(\gamma - \alpha)^2 : rc'(\alpha - \beta)^2,$$

then the equation $\frac{l}{a} + \frac{m}{b} + \frac{n}{c} = 0$ takes the form

$$p(\beta - \gamma)(\alpha - \delta) + q(\gamma - \alpha)(\beta - \delta) + r(\alpha - \beta)(\gamma - \delta) = 0,$$

and the four forms of the equation are found to be

$$\begin{aligned} & \left(\sqrt{p(\beta - \gamma)}, \sqrt{q(\gamma - \alpha)}, \sqrt{r(\alpha - \beta)} \right) (\sqrt{a'U}, \sqrt{b'V}, \sqrt{c'W}, \sqrt{d'T}) = 0, \\ & \left(\sqrt{p(\beta - \gamma)}, \sqrt{r(\beta - \delta)}, \sqrt{q(\gamma - \delta)} \right) (\sqrt{a'U}, \sqrt{b'V}, \sqrt{d'T}, \sqrt{c'W}) = 0, \\ & \left(\sqrt{q(\alpha - \gamma)}, \sqrt{r(\alpha - \delta)}, \sqrt{p(\gamma - \delta)} \right) (\sqrt{a'U}, \sqrt{c'W}, \sqrt{d'T}, \sqrt{b'V}) = 0, \\ & \left(\sqrt{r(\beta - \alpha)}, \sqrt{q(\delta - \alpha)}, \sqrt{p(\beta - \delta)} \right) (\sqrt{b'V}, \sqrt{c'W}, \sqrt{d'T}, \sqrt{a'U}) = 0, \end{aligned}$$

viz., these are the equivalent forms of the original equation assumed to be

$$(\beta - \gamma)\sqrt{pa'U} + (\gamma - \alpha)\sqrt{qb'V} + (\alpha - \beta)\sqrt{rc'W} = 0.$$

Article 50. I remark that the theorem of the variable zomal may be obtained as a transformation theorem: viz., comparing the equation $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$ with the equation $\sqrt{lx} + \sqrt{my} + \sqrt{nz} = 0$; this last belongs to a conic touched by the three lines $x = 0, y = 0, z = 0$; the equation of the same conic must, it is clear, be expressible in a similar form by means of any other three tangents thereof, but the equation of any tangent of the conic is $ax + by + cz = 0$, where a, b, c are any quantities satisfying the condition $\frac{l}{a} + \frac{m}{b} + \frac{n}{c} = 0$; whence, writing $ax + by + cz + dw = 0$, we may introduce w along with any two of the original zomals $x = 0, y = 0, z = 0$, or, instead of them, any three functions of the form w ; and then the mere change of x, y, z, w into U, V, W, T gives the theorem. But it is as easy to conduct the analysis with (U, V, W, T) as with (x, y, z, w) , and, so conducted, it is really the same analysis as that whereby the theorem is established ante, No. 47.

Article 51. It is worth while to exhibit the equation of the curve

$$\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0,$$

in a form containing three new zomals. Observe that the equation $\frac{l}{a} + \frac{m}{b} + \frac{n}{c} = 0$ is satisfied by $a = l\phi\psi, b = m\phi\chi, c = n\psi\chi$, if only $\frac{1}{\phi} + \frac{1}{\psi} + \frac{1}{\chi} = 0$; or say, if $\phi = \alpha' - \alpha'', \psi = \alpha'' - \alpha, \chi = \alpha - \alpha'$. The equation

$$\lambda\sqrt{(\alpha - \alpha')(\alpha - \alpha'')lU} + \mu\sqrt{(\alpha' - \alpha'')(\alpha' - \alpha)mV} + \nu\sqrt{(\alpha'' - \alpha)(\alpha'' - \alpha')nW} = 0$$

is consequently an equation involving three zomals of the proper form; and we can determine λ, μ, ν in suchwise as to identify this with the original equation $\sqrt{lU} + \sqrt{mV} + \sqrt{nW}$, viz., writing successively $U = 0, V = 0, W = 0$, we find

$$\begin{aligned}(\alpha' - \alpha'')\lambda + (\beta' - \beta'')\mu + (\gamma' - \gamma'')\nu &= 0, \\(\alpha'' - \alpha)\lambda + (\beta'' - \beta)\mu + (\gamma'' - \gamma)\nu &= 0, \\(\alpha - \alpha')\lambda + (\beta - \beta')\mu + (\gamma - \gamma')\nu &= 0,\end{aligned}$$

equations which are, as they should be, equivalent to two equations only, and which give

$$\lambda : \mu : \nu = \begin{vmatrix} 1 & 1 & 1 \\ \beta & \beta' & \beta'' \\ \gamma & \gamma' & \gamma'' \end{vmatrix} : \begin{vmatrix} 1 & 1 & 1 \\ \gamma & \gamma' & \gamma'' \\ \alpha & \alpha' & \alpha'' \end{vmatrix} : \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \alpha' & \alpha'' \\ \beta & \beta' & \beta'' \end{vmatrix},$$

and the equation, with these values of λ, μ, ν substituted therein, is in fact the equation of the trizomal curve

$$\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$$

in terms of three new zomals.

It is easy to return to the forms involving one new zomal and any two of the original three zomals.

52. Remark as to the Tetrizomal Curve.

Article 52. I return for a moment to the case of the tetrizomal curve, in order to show that there is not, in regard to it in general, any theorem such as that of the variable zomal. Considering the form $\sqrt{lx} + \sqrt{my} + \sqrt{nz} + \sqrt{pw} = 0$ (the coordinates x, y, z, w are of course connected by a linear equation, but nothing turns upon this), the curve is here a quartic touched twice by each of the lines $x = 0, y = 0, z = 0, w = 0$ (viz., each of these is a double tangent of the curve), and having besides the three nodes $(x = y, z = w), (x = z, y = w), (x = w, y = z)$. But a quartic curve with three nodes, or trinodal quartic, has only four double tangents: that is, besides the lines $x = 0, y = 0, z = 0, w = 0$, there is no line $\alpha x + \beta y + \gamma z + \delta w = 0$ which is a double tangent of the curve; and writing U, V, W, T in place of x, y, z, w , then if U, V, W, T are connected by a linear equation (and, *a fortiori*, if they are not so connected), there is not any curve $\alpha U + \beta V + \gamma W + \delta T = 0$ which is related to the curve in the same way with the lines $U = 0, V = 0, W = 0, T = 0$; or say there is not (besides the curves $U = 0, V = 0, W = 0, T = 0$), any other zomal $\alpha U + \beta V + \gamma W + \delta T = 0$, of the tetrizomal curve. The proof does not show that for special forms of U, V, W, T there may not be zomals, not of the above form $\alpha U + \beta V + \gamma W + \delta T = 0$, but belonging to a separate system. An instance of this will be mentioned in the sequel.

53 to 56. The Theorem of the Variable Zomal of a Trizomal Curve resumed.

Article 53. I resume the foregoing theorem of the variable zomal of the trizomal curve $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$. The variable zomal $T = 0$ is the curve $aU + bV + cW = 0$, where a, b, c are connected by the equation $\frac{l}{a} + \frac{m}{b} + \frac{n}{c} = 0$; that is, it belongs to a single series of curves selected in a certain manner out of the double series $aU + bV + cW = 0$ (a double series, as containing the two variable parameters $a : b : c$). These are the whole series of curves in involution with the given curves $U = 0, V = 0, W = 0$, or being such that the

Jacobian of any three of them is identical with the Jacobian of the three given curves; in particular, the Jacobian of any one of the curves $aU + bV + cW = 0$, and of two of the three given curves, is identical with the Jacobian of the three given curves. I call to mind that, by the Jacobian of the curves $U = 0$, $V = 0$, $W = 0$, is meant the curve

$$J(U, V, W) = \begin{vmatrix} \partial_x U & \partial_y U & \partial_z U \\ \partial_x V & \partial_y V & \partial_z V \\ \partial_x W & \partial_y W & \partial_z W \end{vmatrix} = 0,$$

viz., the curve obtained by equating to zero the Jacobian or functional determinant of the functions U , V , W . Some properties of the Jacobian, which are material as to what follows, are mentioned in the Annex No. I.

For the complete statement of the theorem of the variable zomal, it would be necessary to interpret geometrically the condition $\frac{l}{a} + \frac{m}{b} + \frac{n}{c} = 0$, thereby showing how the single series of the variable zomal is selected out of the double series of the curves $aU + bV + cW = 0$ in involution with the given curves. Such a geometrical interpretation of the condition may be sought for as follows, but it is only in a particular case, as afterwards mentioned, that a convenient geometrical interpretation is thereby obtained.

Article 54. Consider the fixed line $\Omega = px + qy + rz = 0$, and let it be proposed to find the locus of the $(r-1)^2$ pole of the line $\Omega = 0$ in regard to the series of curves $aU + bV + cW = 0$, where $\frac{l}{a} + \frac{m}{b} + \frac{n}{c} = 0$. Take (x, y, z) as the coordinates of any one of the poles in question; then in order that (x, y, z) may belong to one of the $(r-1)^2$ poles of the line $\Omega = px + qy + rz = 0$ in regard to the curve $aU + bV + cW = 0$, we must have

$$\partial_x(aU + bV + cW) : \partial_y(aU + bV + cW) : \partial_z(aU + bV + cW) = p : q : r;$$

or, what is the same thing,

$$a \partial_x U + b \partial_x V + c \partial_x W : a \partial_y U + b \partial_y V + c \partial_y W : a \partial_z U + b \partial_z V + c \partial_z W = p : q : r;$$

and these equations give without difficulty

$$a : b : c = J(V, W, \Omega) : J(W, U, \Omega) : J(U, V, \Omega),$$

whence, substituting in the equation $\frac{l}{a} + \frac{m}{b} + \frac{n}{c} = 0$, we have

$$\frac{l}{J(V, W, \Omega)} + \frac{m}{J(W, U, \Omega)} + \frac{n}{J(U, V, \Omega)} = 0$$

as the equation of the locus in question.

Article 55. In the particular case where $r = 2$, the curves $U = 0$, $V = 0$, $W = 0$ are conics; the $(r-1)^2$ pole is the single pole of the line $\Omega = 0$ in regard to the conic $aU + bV + cW = 0$; and the locus is the curve

$$\frac{l}{J(V, W, \Omega)} + \frac{m}{J(W, U, \Omega)} + \frac{n}{J(U, V, \Omega)} = 0,$$

which is a cubic curve. This cubic curve is in fact the locus of the poles of the line $\Omega = 0$ in regard to the series of conics $aU + bV + cW = 0$, where $\frac{l}{a} + \frac{m}{b} + \frac{n}{c} = 0$.

Article 56. The theorem of the variable zomal may be stated as follows: Given the trizomal curve $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$, if $T = 0$ be any curve of the series $aU + bV + cW = 0$, where $\frac{l}{a} + \frac{m}{b} + \frac{n}{c} = 0$, then the trizomal curve may be expressed by means of the three zomals U, V, T ; or U, W, T ; or V, W, T ; and the equation will be of the form $\sqrt{l'U} + \sqrt{m'V} + \sqrt{n'T} = 0$, where l', m', n' are properly determined. The series of curves $T = 0$ is thus a series of variable zomals of the given trizomal curve.

Part II. (Nos. 57 to 104).—Subsidiary Investigations.

Article 57. We have just been led to consider the conics which pass through two given points. There is no real loss of generality in taking these to be the circular points at infinity, or say the points I, J ; viz., every theorem which in anywise explicitly or implicitly relates to these two points, may, without the necessity of any change in the statement thereof, be understood as a theorem relating instead to any two points P, Q .

I call to mind that a circle is a conic passing through the two points I, J , and that lines at right angles to each other are lines harmonically related to the pair of lines from their intersection to the points I, J respectively, so that when (I, J) are replaced by any two given points whatever, the expression a circle must be understood to mean a conic passing through the two given points; and in speaking of lines at right angles to each other, it must be understood that we mean lines harmonically related to the pair of lines from their intersection to the two given points respectively.

For instance, the theorem that the Jacobian of any three circles is their orthotomic circle, will mean that the Jacobian of any three conics which each of them passes through the two given points is the orthotomic conic through the same two points, that is, the conic such that at each of its intersections with any one of the three conics, the two tangents are harmonically related to the pair of lines from this intersection to the two given points respectively. Such extended interpretation of any theorem is applicable even to the theorems which involve distances or angles; viz., the terms “distance” and “angle” have a determinate signification when interpreted in reference (not to the circular points at infinity, but instead thereof) to any two given points whatever (see as to this my “Sixth Memoir on Quantics,” Nos. 220, *et seq.*, *Phil. Trans.*, vol. CXLIX. (1859), pp. 61–90; see p. 86 [158]). And this being so, the theorem can, without change in the statement thereof, be understood as referring to the two given points.

Article 58. I say then that any theorem (referring explicitly or implicitly) to the circular points at infinity I, J , may be understood as a theorem referring instead to any two given points. We might of course give the theorems in the first instance in terms explicitly referring to the two given points (viz., instead of a circle, speak of a conic through the two given points, and so in other instances); but, as just explained, this is not really more general, and the theorems would be given in a less concise and familiar form. It would not, on the face of the investigations, be apparent that in treating of the polyzomal curves

$$\sqrt{l(\Theta + \lambda\Phi)} + \sqrt{m(\Theta + \mu\Phi)} + \&c. = 0,$$

(Θ = a conic, Φ = a line, as above), that we were really treating of the curves the zomals whereof are circles, and therein of the theories of foci and foci-foci as about to be explained. And for these reasons I shall consider the two points $\Theta = 0, \Phi = 0$, to be the circular points at infinity I, J , and in the investigations, &c., make use of the terms circle, right angles, &c., which, in their ordinary significations, have implicit reference to these two points.

The present Part does not explicitly relate to the theory of polyzomal curves, but contains a series of researches, partly analytical and partly geometrical, which will be made use of in the following Parts III. and IV. of the Memoir.

59 to 62. The Circular Points at Infinity; Rectangular and Circular Coordinates.

Article 59. The coordinates made use of (except in the cases where the general trilinear coordinates (x, y, z) , or any other coordinates, are explicitly referred to), will be either the ordinary rectangular coordinates x, y , or else, as we may term them, the circular coordinates ξ, η ($= x + iy, x - iy$ respectively, $i = \sqrt{-1}$ as usual), but in either case I shall introduce for homogeneity the coordinate z , it being understood that this coordinate is in fact $= 1$, and that it may be retained or replaced by this its value, in different investigations or stages of the same investigation, as may for the time being be most convenient. In more concise terms, we may say that the coordinates are either the rectangular coordinates x, y , and z ($= 1$), or else the circular coordinates ξ, η , and z ($= 1$). The equation of the line infinity is $z = 0$; the points I, J are given by the equations $(x + iy = 0, z = 0)$ and $(x - iy = 0, z = 0)$, or, what is the same thing, by the equations $(\xi = 0, z = 0)$ and $(\eta = 0, z = 0)$ respectively; or in the rectangular coordinates the coordinates of these points are $(-i, 1, 0)$ and $(i, 1, 0)$ respectively, and in the circular coordinates they are $(1, 0, 0)$ and $(0, 1, 0)$ respectively. It is, of course, only for points at infinity that the coordinate z is $= 0$ (and observe that for any such point the x and y or ξ and η coordinates may be regarded as finite); for every point whatever not at infinity the coordinate z is, as stated above, $= 1$.

Article 60. Consider a point A , whose coordinates (rectangular) are $(a, a', 1)$ and (circular) $(\alpha, \alpha', 1)$, viz., $\alpha = a + a'i, \alpha' = a - a'i$; then the equations of the lines through A to the points I, J , are

$$x - az + i(y - a'z) = 0, \quad x - az - i(y - a'z) = 0$$

respectively, or they are

$$\xi - \alpha z = 0, \quad \eta - \alpha' z = 0$$

respectively. These equations, if (a, a') or (α, α') are arbitrary, will, it is clear, be the equations of any two lines through the points I, J , respectively.

Article 61. We have from either of the equations in (x, y, z)

$$(x - az)^2 + (y - a'z)^2 = 0,$$

that is, the distance from each other of any two points $(x, y, 1)$, and $(a, a', 1)$ in a line through I or J is $= 0$. And in particular, if $z = 0$, then $x^2 + y^2 = 0$; that is, the distance of the point $(a, a', 1)$ from I or J is in each case $= 0$.

Article 62. Consider for a moment any three points P, Q, A ; the perpendicular distance of P from QA is $= 2\Delta PQA \div \text{distance } QA$; if Q be any point on the line through A to either of the points I, J , and in particular if Q be either of the points I, J , then the triangle PQA is finite, but the distance QA is $= 0$: that is, the perpendicular distance of P from the line through A to either of the points I, J , that is, from any line through either of these points, is $= \infty$. But, as just stated, the triangle PQA is finite, or say the triangles PIA, PJA are each finite; viz., the coordinates (rectangular) of P, A being $(x, y, z = 1)$, $(a, a', 1)$ or (circular) $(\xi, \eta, z = 1)$, $(\alpha, \alpha', 1)$, the expressions for the doubles of these triangles respectively are

$$\begin{vmatrix} x & y & z \\ -i & 1 & 0 \\ a & a' & 1 \end{vmatrix}, \quad \begin{vmatrix} x & y & z \\ i & 1 & 0 \\ a & a' & 1 \end{vmatrix},$$

that is, they are (rectangular coordinates) $x - az + i(y - a'z)$, $x - az - i(y - a'z)$, or (circular coordinates) $\xi - \alpha z$, $\eta - \alpha' z$.

Representing the double areas by $\triangle PIA$, $\triangle PJA$, respectively, and the squared distance of the points A , P , by $\mathfrak{A}$, we have

$$\mathfrak{A} = (x - az)^2 + (y - a'z)^2 = (\xi - \alpha z)(\eta - \alpha' z) = \triangle PIA \cdot \triangle PJA.$$

63. Antipoints; Definition and Fundamental Properties.

Article 63. Two pairs of points (A, B) and (A_1, B_1) which are such that the lines AB , A_1B_1 bisect each other at right angles in a point O in such wise that $OA = OB = iOA_1 = iOB_1$, are said to be *antipoints*, each of the other. In rectangular coordinates, taking the coordinates of (A, B) to be $(a, 0, 1)$ and $(-a, 0, 1)$, those of (A_1, B_1) will be $(0, ai, 1)$ and $(0, -ai, 1)$ respectively, whence joining the points (A, B) with the points (I, J) , the points A_1, B_1 , are given as the intersections of the lines AI and BJ , and of the lines AJ and BI respectively. Or, what is the same thing, in any quadrilateral wherein I, J are opposite angles, the remaining pairs (A, B) and (A_1, B_1) are antipoints each of the other.

Article 64. In circular coordinates, if the coordinates of A are $(\alpha, \alpha', 1)$, and those of B are $(\beta, \beta', 1)$, then the equations of

$$\begin{aligned} AI, \quad AJ \text{ are } \xi - \alpha z = 0, \quad \eta - \alpha' z = 0, \\ BI, \quad BJ \text{ are } \xi - \beta z = 0, \quad \eta - \beta' z = 0, \end{aligned}$$

whence the equations of

$$\begin{aligned} A_1I, \quad A_1J \text{ are } \xi - \alpha_1 z = 0, \quad \eta - \beta' z = 0, \\ B_1I, \quad B_1J \text{ are } \xi - \beta z = 0, \quad \eta - \alpha' z = 0. \end{aligned}$$

Article 65. Considering any point P the coordinates of which are $\xi, \eta, z (= 1)$, let $\mathfrak{A}, \mathfrak{B}, \mathfrak{A}_1, \mathfrak{B}_1$ be its squared distances from the points A, B, A_1, B_1 respectively; then by what precedes

$$\begin{aligned} \mathfrak{A} &= (\xi - \alpha z)(\eta - \alpha' z), \\ \mathfrak{B} &= (\xi - \beta z)(\eta - \beta' z), \\ \mathfrak{A}_1 &= (\xi - \alpha_1 z)(\eta - \beta' z), \\ \mathfrak{B}_1 &= (\xi - \beta z)(\eta - \alpha' z), \end{aligned}$$

and thence

$$\mathfrak{A} \cdot \mathfrak{B} = \mathfrak{A}_1 \cdot \mathfrak{B}_1;$$

that is, *the product of the squared distances of a point from any two points A, B , is equal to the product of the squared distances of the same point from the two antipoints A_1, B_1* . This theorem, which was, I believe, first given by me in the *Educational Times* (see reprint, vol. vi. 1866, p. 81), is an important one in the theory of foci. It is to be further noticed that we have

$$\mathfrak{A} + \mathfrak{B} - \mathfrak{A}_1 - \mathfrak{B}_1 = (\alpha - \beta)(\alpha' - \beta')z^2 = Kz^2 = K,$$

if $K = (\alpha - \alpha')(\beta - \beta')$, be the squared distance of the points A, B , = squared distance of points A_1, B_1 .

66. Antipoints of a Circle.

Article 66. A similar notion to that of two pairs of antipoints is as follows, viz., if from the centre of a circle perpendicular to its plane and in opposite senses, we measure off two distances each $= i$ into the radius, the extremities of these distances are *antipoints of the circle*. It is clear that the antipoints of the circle and the extremities of any diameter thereof are (in the plane of these four points) pairs of antipoints. It is to be added that each antipoint is the centre of a sphere radius zero, or say of a cone-sphere, passing through the circle: the circle is thus the intersection of the two cone-spheres having their centres at the two antipoints respectively.

67. Antipoints in relation to a Pair of Orthotomic Circles.

Article 67. It is a well-known property that if any circle pass through the points (A, B) , and any other circle through the antipoints (A_1, B_1) , then these two circles cut at right angles. Conversely if a circle pass through the points A, B , then all the orthotomic circles which have their centres on the circle through the antipoints (A_1, B_1) will pass through the points A, B ; and the like as to any circle through A_1, B_1 .

72 to 77. On a System of Sixteen Points.

Article 68. Take (A, B, C, D) any four concyclic points, and let the antipoints of

$$\begin{aligned} (B, C), \quad (A, D) &\text{ be } (B_1, C_1), \quad (A_1, D_1), \\ (C, A), \quad (B, D) &\text{ be } (C_2, A_2), \quad (B_2, D_2), \\ (A, B), \quad (C, D) &\text{ be } (A_3, B_3), \quad (C_3, D_3), \end{aligned}$$

then each of the three new sets (A_1, B_1, C_1, D_1) , (A_2, B_2, C_2, D_2) , (A_3, B_3, C_3, D_3) will be a set of four concyclic points.

Article 69. Let O be the centre of the circle through (A, B, C, D) , say of the circle O , and then, the lines BC, AD meeting in R , the lines CA, BD in S , and the lines AB, CD in T , let each of these points be made the centre of a circle orthotomic to O , viz., let these new circles be called the circles R, S, T respectively.

As regards the circle R , since its centre lies in BC , the circle passes through (B_1, C_1) ; and since the centre lies in AD , the circle passes through (A_1, D_1) ; that is, the four points (A_1, B_1, C_1, D_1) lie in the circle R . Similarly (A_2, B_2, C_2, D_2) lie in the circle S , and (A_3, B_3, C_3, D_3) in the circle T .

Article 70. The points R, S, T are conjugate points in relation to the circle O ; that is, ST, TR, RS are the polars of R, S, T respectively in regard to this circle; and they are, consequently, at right angles to the lines OR, OS, OT respectively; viz., the four centres O, R, S, T are such that the line joining any two of them cuts at right angles the line joining the other two of them, and we see that the relation between the four sets is in fact a symmetrical one; this is most easily seen by consideration of the circular points at infinity

I, J ; the four sets of points may be arranged thus

	I	I	I	I
J	A	A_1	A_2	A_3
J	B	B_1	B_2	B_3
J	C	C_1	C_2	C_3
J	D	D_1	D_2	D_3

in such wise that any four of them in the same vertical line pass through I , and any four in the same horizontal line pass through J ; and this being so, starting for instance with (A_1, B_1, C_1, D_1) we have antipoints

$$\begin{aligned} &\text{of } (A_1, C_1), \quad (A_1, B_1) \text{ are } (A, C), \quad (A_3, B_3), \\ &\text{of } (A_3, B_3), \quad (A, D) \text{ are } (C_2, D_2), \quad (A_2, D_2), \\ &\text{of } (A_2, D_2), \quad (C, D) \text{ are } (A, B), \quad (C, D), \end{aligned}$$

and similarly if we start from (A_1, B_2, C_2, D_2) or (A_3, B_3, C_3, D_3) .

Article 71. I return for a moment to the construction of (A_1, B_1, C_1, D_1) ; these are points on the circle R , and (A_1, D_1) are the antipoints of (B, C) ; that is, they are the intersections of the circle R by the line at right angles to BC from its middle point, or, what is the same thing, by the perpendicular on BC from O . Similarly (A_1, D_1) are the antipoints of (A, D) ; that is, they are the intersections of the circle R by the perpendicular on AD from O . And the like as to (A_2, B_2, C_2, D_2) and (A_3, B_3, C_3, D_3) respectively.

Article 72. Hence, starting with the points A, B, C, D on the circle O , and constructing as above the circles R, S, T , and constructing also the perpendiculars from O on the six chords AB, AC , &c.,

$$\begin{array}{llll} \text{the perpendiculars on } BC, & \text{meet circle } R \text{ in } & (B_1, C_1), & (A_1, D_1), \\ CA, & BD & S & (C_2, A_2), \quad (B_2, D_2), \\ AB, & CD & T & (A_3, B_3), \quad (C_3, D_3), \end{array}$$

so that the whole system is given by means of the circles R, S, T , and the six perpendiculars.

Article 73. If to fix the ideas (A, B, C, D) are real points taken in order on the real circle O , then the points R, S, T are each of them real; but R and T lie outside, S inside the circle O . The circles R and T are consequently real, but the circle S imaginary, viz., its radius is $= i$ into a real quantity; the imaginary points (A_2, B_2, C_2, D_2) are thus given as the intersections of a real circle by a pair of real lines, and the like as to the imaginary points (A_3, B_3, C_3, D_3) ; but the imaginary points (A_1, B_1, C_1, D_1) are only given as the intersections of an imaginary circle (centre real and radius a pure imaginary) by a pair of real lines. The points (C_2, A_2) qua antipoints of (C, A) are easily constructed as the intersections of a real circle by a real line, and the like as to the points (B_2, D_2) qua antipoints of (B, D) , but the construction for the two pairs of points cannot be effected by means of the same real circle.

78 to 80. Property in regard to Four Confocal Conics.

Article 74. All the conics which pass through the four concyclic points A, B, C, D , have their axes in fixed directions; but three such conics are the line-pairs $(BC, AD), (CA, BD),$

and (AB, CD) , whence the directions of the axes are those of the bisectors of the angles formed by any one of these pairs of lines; hence, in particular, considering either axis of a conic through the four points, the lines AB and CD are equally inclined on opposite sides to this axis, and this leads to the theorem that the antipoints (A_3, B_3) , (C_3, D_3) are in a conic confocal to the given conic through (A, B, C, D) ; whence, also, considering any given conic whatever through (A, B, C, D) , the points (A_1, B_1, C_1, D_1) , (A_2, B_2, C_2, D_2) , (A_3, B_3, C_3, D_3) lie severally in three conics, each of them confocal with the given conic.

Article 75. To prove this, consider any two confocal conics, say an ellipse and a hyperbola, and let F be one of their four intersections; join F with the common centre O , and let OT , ON be parallel to the tangent and normal respectively of the ellipse at the point F . OF , OT are in direction conjugate axes of the ellipse, and OF , ON are in direction conjugate axes of the hyperbola; and if they are also the axes in magnitude, that is, if the points T , N are the intersections of OT with the ellipse and of ON with the hyperbola respectively, then it is easy to show that $OT^2 + ON^2 = 0$. And this being so, imagine on the ellipse any two points A , B such that the chord AB is parallel to OT , that is conjugate to OF ; AB is bisected by OF , say in a point K , or we have parallel to OT the semichords or ordinates $KA = KB$; and we may, perpendicularly to this or parallel to ON , draw through K in the hyperbola a chord A_3B_3 , which chord will be bisected in K , or we shall have $KA_3 = KB_3$. Hence KA , KA_3 are in the ellipse and the hyperbola respectively ordinates conjugate to the same diameter OF , and the semi-diameters conjugate to OF being OT , ON respectively, we have $KA^2 (= KB^2) : KA_3^2 (= KB_3^2) = OT^2 : ON^2$; this is, $KA^2 = KB^2 = -KA_3^2 = -KB_3^2$; or (A_3, B_3) will be the antipoints of (A, B) .

Article 76. Conversely, if in the ellipse we have the two points (A, B) , then drawing the diameter OF conjugate to AB , and through its extremity F the confocal hyperbola, then the antipoints (A_3, B_3) will lie on the hyperbola. And similarly, if on the ellipse we have the two points (C, D) , then drawing the diameter OG conjugate to CD , and through its extremity G a confocal hyperbola, the antipoints (C_3, D_3) will lie on the hyperbola. Suppose (A, B, C, D) are concyclic, then, as noticed, AB and CD will be equally inclined on opposite sides to the transverse axis of the ellipse; the conjugate diameters OF , OG will therefore be equally inclined on opposite sides of the transverse axis; and the points F and G will therefore be situate symmetrically on opposite sides of the transverse axis, that is, the points F and G will respectively determine the same confocal hyperbola, and we have thus the required theorem, viz., if (A, B, C, D) are any four concyclic points on an ellipse, or say on a conic, and if (A_3, B_3) are the antipoints of (A, B) , and (C_3, D_3) the antipoints of (C, D) , then (A_3, B_3, C_3, D_3) will lie on a conic confocal with the given conic.

81 to 85. System of the Sixteen Points, the Axial Case.

Article 77. The theorems hold good when the four points A , B , C , D are in a line; the antipoints (B_1, C_1) of (B, C) , &c., are in this case situate symmetrically on opposite sides of the line, so that it is evident at sight that we have (A_1, B_1, C_1, D_1) , (A_2, B_2, C_2, D_2) , (A_3, B_3, C_3, D_3) , each set in a circle; and that the centres R , S , T of these circles lie in the line. The construction for the general case becomes, however, indeterminate, and must therefore be varied. If in the general case we take any circle through (B, C) , and any circle through (A, D) , then the circle R cuts at right angles these two circles, and has, consequently, its centre in the radical axis of the two circles; whence, when the four points are in a line, taking any circle through (B, C) , or in particular the circle on BC as diameter, and any

circle through (A, D) , or in particular the circle on AD as diameter, the radical axis of these two circles intersects the line in the required centre R , and the circle R is the circle with this centre cutting at right angles the two circles respectively; the circles S and T are, of course, obtained by the like construction in regard to the combinations $(C, A; B, D)$ and $(A, B; C, D)$, respectively. It may be added, that we have

$R \left\{ \begin{array}{l} \text{centre} \\ \text{extremities} \end{array} \right. \begin{array}{l} \\ BC; \quad AD \end{array}$	$S \left\{ \begin{array}{l} \text{centre} \\ \text{sibiconjugate points of involutions} \end{array} \begin{array}{l} \\ CA; \quad BD \end{array}$	$T \left\{ \begin{array}{l} \text{centre} \\ \text{of circles} \end{array} \begin{array}{l} \\ AB; \quad CD \end{array}$
---	---	--

and that (as in the general case) the circles R, S, T intersect each pair of them at right angles; and they are evidently each intersected at right angles by the line $ABCD$ (or axis of the figure), which replaces the circle O in the general case.

Article 78. If the points A, B, C, D are taken in order on the line, then the points R, S, T are all real, viz., the point R is situate, on one side or the other, outside AD , but the points S and T are each of them situate between B and C ; the circles R and T are real, but the circle S has its radius a pure imaginary quantity.

Article 79. If one of the four points, suppose D , is at infinity on the line, then the antipoints of (A, D) , of (B, D) , and of (C, D) are each of them the two points (I, J) .

It would at first sight appear that the only conditions for the circles R, S, T were the conditions of passing through the antipoints of (B, C) , of (C, A) , and of (A, B) respectively, and that these circles thus became indeterminate; but in fact the definition of the circles is then as follows, viz., R has its centre at A , and passes through the antipoints of (B, C) : (whence square of radius = $AB \cdot AC$); S has its centre at B , and passes through the antipoints of (C, A) (whence square of radius = $BC \cdot BA$); T has its centre at C , and passes through the antipoints of (A, B) (whence square of radius = $CA \cdot CB$); and we have thus three circles each intersecting the other two at right angles.

Article 80. If two of the points, say C, D , are at infinity, then the circle R has its centre at the middle point of AB , and passes through I and J , that is, it is the circle on AB as diameter; similarly S is the line through B at right angles to AB , and T is the line through A at right angles to AB ; the system of three circles is thus a system of a circle and two lines at right angles to each other and each passing through the extremity of a diameter of the circle.

Article 81. If three of the points, say B, C, D , are at infinity, then A is any point, R is any circle through A , S is the line through A at right angles to OA , and T is the line through A at right angles to OA ; but this is a system which does not present any thing worthy of notice.

86 to 91. The Involution of Four Circles.

Article 82. Given any four circles, there are in general two circles which cut each of the four given circles at right angles; these two circles are called the *orthotomic circles* of the four given circles. If the four given circles have a common orthotomic circle, then they form a system of circles in involution.

Article 83. The condition that four circles may have a common orthotomic circle may be expressed analytically as follows: Let the equations of the four circles be

$$S_1 = x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0, \quad \&c.,$$

then the equation of a circle cutting all four at right angles is

$$S - \lambda_1 S_1 - \lambda_2 S_2 - \lambda_3 S_3 - \lambda_4 S_4 = 0,$$

where $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are determined by the conditions that the coefficients of x and y in S vanish, and that the radius of S is zero. These conditions give four equations, which in general cannot be satisfied; but if the four circles satisfy a certain condition, the equations become compatible, and there is a circle cutting all four at right angles.

Article 84. The condition may also be expressed geometrically: If four circles have a common orthotomic circle, then the centres of the four circles lie on a circle, and the squares of the radii are proportional to the squares of the distances of the centres from a fixed point.

Article 85. In the case where the four circles pass through two common points, they have in general a common orthotomic circle; for if A, B are the two common points, then any circle through the antipoints of (A, B) cuts all four circles at right angles.

Article 86. The system of four circles in involution is connected with the theory of the sixteen points as follows: Given four concyclic points (A, B, C, D) , the circles through any three of these points form a system of four circles in involution; and the orthotomic circle of this system is the circle through the antipoints of any pair of the four points.

Article 87. The involution of four circles may be generalised as follows: Given any four curves of the second order passing through two fixed points, these curves form a system in involution, and there exists a curve of the second order cutting all four at right angles, that is, such that at each intersection the tangents to the two curves are harmonically related to the lines joining the intersection to the two fixed points.

92. On a Locus connected with the foregoing Properties.

Article 88. If, as above, A, B, C, D are any four points, and $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$ are the squared distances of a current point P from the four points respectively, then the locus of the foci of the conics which pass through the four points is the tetrazomal curve

$$a\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}} + d\sqrt{\mathfrak{D}} = 0.$$

In fact the sum $a\mathfrak{A} + b\mathfrak{B} + c\mathfrak{C} + d\mathfrak{D}$ has, it has been seen, a constant value for all positions of the point P ; taking Q to be the other focus, its squared distances are $(k - \sqrt{\mathfrak{A}})^2$, &c., whence for the first-mentioned focus we have

$$a\mathfrak{A} + b\mathfrak{B} + c\mathfrak{C} + d\mathfrak{D} = a(k - \sqrt{\mathfrak{A}})^2 + b(k - \sqrt{\mathfrak{B}})^2 + c(k - \sqrt{\mathfrak{C}})^2 + d(k - \sqrt{\mathfrak{D}})^2,$$

or recollecting that $a + b + c + d = 0$, it follows that we have for the locus in question

$$a\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}} + d\sqrt{\mathfrak{D}} = 0;$$

this locus will be discussed in the sequel. I remark here, that in the case where the four points are on a circle, then (as mentioned above), the axes of the several conics are in the same fixed directions; there are thus two sets of foci, those on the axis in one direction, and those on the axis in the other direction; it might therefore be anticipated, and it will appear, that in this case the tetrazomal breaks up into two trizomal curves.

93 to 98. Formulae as to the two Sets (A, B, C, D) and (A_1, B_1, C_1, D_1) , each of four Concyelic Points.

Article 89. Consider the four points A, B, C, D on a circle, then taking, as before, their circular coordinates to be $(\alpha, \alpha', 1), (\beta, \beta', 1), (\gamma, \gamma', 1), (\delta, \delta', 1)$, the condition that the points may be on a circle is

$$\begin{vmatrix} 1 & \alpha & \alpha' & \alpha\alpha' \\ 1 & \beta & \beta' & \beta\beta' \\ 1 & \gamma & \gamma' & \gamma\gamma' \\ 1 & \delta & \delta' & \delta\delta' \end{vmatrix} = 0,$$

viz., this equation may be written

$$(\beta - \gamma)(\alpha - \delta) : (\gamma - \alpha)(\beta - \delta) : (\alpha - \beta)(\gamma - \delta) = (\beta' - \gamma')(\alpha' - \delta') : (\gamma' - \alpha')(\beta' - \delta') : (\alpha' - \beta')(\gamma' - \delta')$$

or if, for shortness, we take

$$\begin{aligned} a &= \beta - \gamma, & f &= \alpha - \delta, & a' &= \beta' - \gamma', & f' &= \alpha' - \delta', \\ b &= \gamma - \alpha, & g &= \beta - \delta, & b' &= \gamma' - \alpha', & g' &= \beta' - \delta', \\ c &= \alpha - \beta, & h &= \gamma - \delta, & c' &= \alpha' - \beta', & h' &= \gamma' - \delta', \end{aligned}$$

and consequently

$$\begin{aligned} af + bg + ch &= 0, \\ a'f' + b'g' + c'h' &= 0, \end{aligned}$$

$$\begin{aligned} a &= g - h, & a' &= g' - h', \\ b &= h - f, & b' &= h' - f', \\ c &= f - g, & c' &= f' - g', \end{aligned}$$

$$a + b + c = 0, \quad a' + b' + c' = 0,$$

then the equation is

$$af : bg : ch = a'f' : b'g' : c'h'.$$

Article 90. Let a, b, c, d denote as before ($a : b : c : d = BCD : -CDA : DAB : -ABC$), then we have

$$a : b : c : d = \begin{vmatrix} \beta & \beta' & 1 \\ \gamma & \gamma' & 1 \\ \delta & \delta' & 1 \end{vmatrix} : - \begin{vmatrix} \gamma & \gamma' & 1 \\ \delta & \delta' & 1 \\ \alpha & \alpha' & 1 \end{vmatrix} : \begin{vmatrix} \delta & \delta' & 1 \\ \alpha & \alpha' & 1 \\ \beta & \beta' & 1 \end{vmatrix} : - \begin{vmatrix} \alpha & \alpha' & 1 \\ \beta & \beta' & 1 \\ \gamma & \gamma' & 1 \end{vmatrix},$$

and we may write

$$\begin{aligned} a &= ah' - a'h, & a &= a'g - ag', & g &= gh' - g'h, \\ b &= bh' - b'h, & b &= b'f - bf', & h &= hf' - h'f, \\ c &= ch' - c'h, & c &= cf' - c'f, & f &= fg' - f'g, \\ d &= -(cb' - c'b), & d &= -(ac' - a'c), & d &= -(ba' - b'a), \end{aligned}$$

viz., the expressions in the same horizontal line are equal, and a, b, c, d are proportional to the expressions in the four lines respectively.

Article 91. I say that we have

$$\frac{a}{a'} = \frac{b}{b'} = \frac{c}{c'} = -\frac{d}{d'},$$

viz., this will be the case if

$$bc' = fg'd, \quad ac'h = hf'd, \quad a'bc = fg'd,$$

and selecting the convenient expressions for a, b, c, d , these equations become

$$bc'(gh' - g'h) = g'h(cb' - c'b), \quad ac'(hf' - h'f) = fh'(ac' - a'c), \quad a'b(fg' - f'g) = fg'(ba' - b'a),$$

viz., these equations are respectively $bgc'h' = b'g'ch, cha'f' = c'h'af, afb'g' = a'f'bg$, and are consequently satisfied. It thus appears that the equation

$$\frac{l}{a} + \frac{m}{b} + \frac{n}{c} + \frac{p}{d} = 0$$

is transformable into

$$\frac{l}{c'f'} + \frac{m}{c'g'} + \frac{n}{a'f'} + \frac{p}{f'g'} = 0,$$

which is of course one of a system of similar forms.

Article 92. Take (A_1, D_1) the antipoints of (A, D) ; (B_1, C_1) the antipoints of (B, C) ; or say that the circular coordinates of A_1, B_1, C_1, D_1 are $(\alpha, \delta', 1), (\beta, \gamma', 1), (\gamma, \beta', 1), (\delta, \alpha', 1)$ respectively; the points A_1, B_1, C_1, D_1 are, as above mentioned, on a circle, the condition that this may be so being in fact

$$\begin{vmatrix} 1 & \alpha & \delta' & \alpha\delta' \\ 1 & \beta & \gamma' & \beta\gamma' \\ 1 & \gamma & \beta' & \gamma\beta' \\ 1 & \delta & \alpha' & \delta\alpha' \end{vmatrix} = 0,$$

equivalent to

$$af : bg : ch = a'f' : b'g' : c'h'.$$

Article 93. Let (a_1, b_1, c_1, d_1) be the corresponding quantities to (a, b, c, d) , viz.,

$$a_1 : b_1 : c_1 : d_1 = B_1C_1D_1 : -C_1D_1A_1 : D_1A_1B_1 : -A_1B_1C_1;$$

we have

$$a_1 : b_1 : c_1 : d_1 = \begin{vmatrix} \beta & \gamma' & 1 \\ \gamma & \beta' & 1 \\ \delta & \alpha' & 1 \end{vmatrix} : - \begin{vmatrix} \gamma & \beta' & 1 \\ \delta & \alpha' & 1 \\ \alpha & \delta' & 1 \end{vmatrix} : \begin{vmatrix} \delta & \alpha' & 1 \\ \alpha & \delta' & 1 \\ \beta & \gamma' & 1 \end{vmatrix} : - \begin{vmatrix} \alpha & \delta' & 1 \\ \beta & \gamma' & 1 \\ \gamma & \beta' & 1 \end{vmatrix},$$

giving rise to a similar set of forms

$$\begin{array}{lll} a_1 = \dots, & -ac' + ba', a'g + b'a, & -c'g - b'h, \\ b_1 = -c'h - g'h, & \dots, -f'b - g'f, & -f'h + c'f, \\ c_1 = b'c + h'g, & -f'c + h'f, \dots, & f'g + g'f, \\ d_1 = g'c + h'h, & -h'a + a'c, -a'b - g'a, & \dots, \end{array}$$

and leading to

$$\frac{a_1}{a'} = \frac{b_1}{c'g'} = \frac{c_1}{a'f'} = -\frac{d_1}{f'g'},$$

so that the equation

$$\frac{l_1}{a_1} + \frac{m_1}{b_1} + \frac{n_1}{c_1} + \frac{p_1}{d_1} = 0,$$

is transformable into

$$\frac{l_1}{c'f'} + \frac{m_1}{c'g'} + \frac{n_1}{a'f'} + \frac{p_1}{f'g'} = 0.$$

Article 94. Let A, B, C, D be, as above, points on a circle; (A_1, D_1) and (B_1, C_1) the antipoints of (A, D) , (B, C) respectively. Write

$$\begin{aligned} \mathfrak{A} &= (\xi - \alpha z)(\eta - \alpha' z), & \mathfrak{A}_1 &= (\xi - \alpha z)(\eta - \delta' z), \\ \mathfrak{B} &= (\xi - \beta z)(\eta - \beta' z), & \mathfrak{B}_1 &= (\xi - \beta z)(\eta - \gamma' z), \\ \mathfrak{C} &= (\xi - \gamma z)(\eta - \gamma' z), & \mathfrak{C}_1 &= (\xi - \gamma z)(\eta - \beta' z), \\ \mathfrak{D} &= (\xi - \delta z)(\eta - \delta' z), & \mathfrak{D}_1 &= (\xi - \delta z)(\eta - \alpha' z); \end{aligned}$$

then we have identically

$$\begin{aligned} f'\mathfrak{B} &= gg'\mathfrak{A} + cc'\mathfrak{D} + gc'\mathfrak{A}_1 + cg'\mathfrak{D}_1, \\ f'\mathfrak{C} &= hh'\mathfrak{A} + bb'\mathfrak{D} - hh'\mathfrak{A}_1 - bh'\mathfrak{D}_1, \\ g'\mathfrak{B}_1 &= gh'\mathfrak{A} - cb'\mathfrak{D} - gb'\mathfrak{A}_1 + ch'\mathfrak{D}_1, \\ g'\mathfrak{C}_1 &= hg'\mathfrak{A} - bc'\mathfrak{D} + hc'\mathfrak{A}_1 - bg'\mathfrak{D}_1, \end{aligned}$$

or, in the foregoing notation,

$$\begin{aligned} f'\mathfrak{B} &= gg'\mathfrak{A} + cc'\mathfrak{D} + gc'\mathfrak{A}_1 + cg'\mathfrak{D}_1, \\ f'\mathfrak{C} &= hh'\mathfrak{A} + bb'\mathfrak{D} - hh'\mathfrak{A}_1 - bh'\mathfrak{D}_1, \\ g'\mathfrak{B}_1 &= gh'\mathfrak{A} - cb'\mathfrak{D} - gb'\mathfrak{A}_1 + ch'\mathfrak{D}_1, \\ g'\mathfrak{C}_1 &= hg'\mathfrak{A} - bc'\mathfrak{D} + hc'\mathfrak{A}_1 - bg'\mathfrak{D}_1. \end{aligned}$$

99 to 104. Further Properties in relation to the same Sets (A, B, C, D) and (A_1, B_1, C_1, D_1) .

Article 95. It is to be shown that in virtue of these equations, and if moreover $\frac{a}{l} + \frac{b}{m} + \frac{c}{n} + \frac{d}{p} = 0$, then it is possible to find l_1, m_1, n_1, p_1 , such that we have identically

$$-l\mathfrak{A} + m\mathfrak{B} + n\mathfrak{C} - p\mathfrak{D} + l_1\mathfrak{A}_1 - m_1\mathfrak{B}_1 - n_1\mathfrak{C}_1 + p_1\mathfrak{D}_1 = 0.$$

This equation will in fact be identically true if only

$$\begin{aligned} -ff'l + gg'm + hh'n & & -gh'm_1 - g'h'n_1 &= 0, \\ cc'm + hh'n - ff'p & & + ci'n_1 + hc'p_1 &= 0, \\ gc'm - hh'n & & + ff'l_1 + gh'n_1 - hc'p_1 &= 0, \\ cg'm - hh'n & & + ch'm_1 + hg'p_1 &+ lf'p_1 = 0. \end{aligned}$$

From the first and second equations eliminating m_1 or n_1 , the other of these quantities disappears of itself, and we thus obtain two equations which must be equivalent to a single one, viz., we have

$$\begin{aligned} bc'ff'l + c'g'af'm + bha'f'n + g'h'f'p &= 0, \\ b'cf'f'l + cga'f'm + h'h'af'n + gh'ff'p &= 0; \end{aligned}$$

which equations may also be written

$$\begin{aligned} \frac{l}{c'f'} + \frac{m}{c'g'} + \frac{n}{a'f'} + \frac{p}{f'g'} &= 0, \\ \frac{l}{a'h'} + \frac{m}{b'h'} + \frac{n}{a'f'} + \frac{p}{a'b'} &= 0, \end{aligned}$$

and it thus appears that the equations are equivalent to each other, and to the assumed relation

$$\frac{a}{l} + \frac{b}{m} + \frac{c}{n} + \frac{d}{p} = 0.$$

Article 96. Similarly, from the third and fourth equations eliminating m_1 or n_1 , the other of these quantities disappears of itself, and we find

$$\begin{aligned} b'c'ff'l_1 - cga'f'm_1 + afc'g'n_1 - dg'f'p_1 &= 0, \\ bKff'l_1 - afb'h'm_1 + hha'f'n_1 - h'hff'p_1 &= 0, \end{aligned}$$

equations which may be written

$$\frac{l_1}{cf} + \frac{m_1}{cg} + \frac{n_1}{af} + \frac{p_1}{fg} = 0, \quad \frac{l_1}{ac} + \frac{m_1}{eg} + \frac{n_1}{af} + \frac{p_1}{ga} = 0,$$

and which are equivalent to each other, and to

$$\frac{a_1}{l_1} + \frac{b_1}{m_1} + \frac{c_1}{n_1} + \frac{d_1}{p_1} = 0.$$

Article 97. It thus appears that we have identically

$$l\mathfrak{A} - m\mathfrak{B} - n\mathfrak{C} + p\mathfrak{D} = l_1\mathfrak{A}_1 - m_1\mathfrak{B}_1 - n_1\mathfrak{C}_1 + p_1\mathfrak{D}_1,$$

and that the condition $\frac{a}{l} + \frac{b}{m} + \frac{c}{n} + \frac{d}{p} = 0$ is equivalent to the condition $\frac{a_1}{l_1} + \frac{b_1}{m_1} + \frac{c_1}{n_1} + \frac{d_1}{p_1} = 0$. The theorem is thus established.

Article 98. It follows from the foregoing that if we have the tetrazomal curve

$$\sqrt{l\mathfrak{A}} + \sqrt{m\mathfrak{B}} + \sqrt{n\mathfrak{C}} + \sqrt{p\mathfrak{D}} = 0,$$

where $\frac{a}{l} + \frac{b}{m} + \frac{c}{n} + \frac{d}{p} = 0$, then this curve is identical with the tetrazomal curve

$$\sqrt{l_1\mathfrak{A}_1} + \sqrt{m_1\mathfrak{B}_1} + \sqrt{n_1\mathfrak{C}_1} + \sqrt{p_1\mathfrak{D}_1} = 0,$$

where $\frac{a_1}{l_1} + \frac{b_1}{m_1} + \frac{c_1}{n_1} + \frac{d_1}{p_1} = 0$; and the two forms of the equation are equivalent to each other.

Article 99. In particular, if A, B, C, D are four concyclic points, and $(A_1, D_1), (B_1, C_1)$ are the antipoints of $(A, D), (B, C)$ respectively, then the tetrazomal curve

$$a\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}} + d\sqrt{\mathfrak{D}} = 0$$

breaks up into two trizomal curves, viz., into

$$\sqrt{gg'\mathfrak{A}} + \sqrt{cc'\mathfrak{D}} + \sqrt{gc'\mathfrak{A}_1} + \sqrt{cg'\mathfrak{D}_1} = 0,$$

and

$$\sqrt{hh'\mathfrak{A}} + \sqrt{bb'\mathfrak{D}} - \sqrt{hh'\mathfrak{A}_1} - \sqrt{bh'\mathfrak{D}_1} = 0;$$

or, what is the same thing, the locus of the foci of conics through four concyclic points breaks up into two trizomal curves.

Part III. (Nos. 105 to 157).—On the Theory of Foci.

105 to 110. Explanation of the General Theory.

Article 100. If from a focus of a conic we draw two tangents to the curve, these pass respectively through the two circular points at infinity, and we have thence the generalised definition of a focus as established by Plücker, viz., in any curve a *focus* is a point such that the lines joining it with the two circular points at infinity are respectively tangents to the curve; or, what is the same thing, if from each of the circular points at infinity, say from the points I, J , tangents are drawn to the curve, the intersections of each tangent from the one point with each tangent from the other point are the foci of the curve. A curve of the class n has thus in general n^2 foci.

It is to be added that, as in the conic the line joining the points of contact of the two tangents from a focus is the directrix corresponding to that focus, so in general the line joining the points of contact of the tangents from the focus through the points I, J respectively is the directrix corresponding to the focus in question.

Article 101. A circular point at infinity I or J , may be an ordinary or a singular point on the curve, and the tangent at this point then counts, or, in the case of a multiple point, the tangents at this point count a certain number of times, say q times, among the tangents which can be drawn to the curve from the point; the number of the remaining tangents is thus $= n - q$. In particular, if the circular point at infinity be an ordinary point, then the tangent counts twice, or we have $q = 2$; if it be a node, each of the tangents counts twice, or $q = 4$; if it be a cusp, the tangent counts three times, or $q = 3$. Similarly, if the other circular point at infinity be an ordinary or a singular point on the curve, the tangent or tangents there count a certain number of times, say q' times, among the tangents to the curve from this point; the number of the remaining tangents is thus $= n - q'$. And if as usual we disregard the tangents at the two points I, J respectively, and attend only to the remaining tangents, the number of the foci is $= (n - q)(n - q')$.

Article 102. Among the tangents from the point I or J there may be a tangent which, either from its being a multiple tangent (that is, a tangent having ordinary contact at two or more distinct points), or from being an osculating tangent at one or more points, counts a certain number of times, say r , among the tangents from the point in question. Similarly, if among the tangents from the other point J or I , there is a tangent which counts r' times, then the foci are made up as follows, viz. we have

Intersections of the two singular tangents counting as rr' foci.

Intersections of the first singular tangent with each of
the ordinary tangents from the other circular point at
infinity, as $(n - q' - r')r$

$$(n - q - r)r'$$

Do. for second singular tangent,

$$(n - q' - r')r$$

Intersections of the ordinary tangents

$$(n - q - r)(n - q' - r')$$

Giving together the $(n - q)(n - q')$ foci

and the like observation applies to the more general case where the tangents from each of the points I, J include more than one singular tangent.

Article 103. There is yet another case to be considered; the line infinity may be an ordinary or a singular tangent to the curve: assuming that it counts s times among the tangents from either of the circular points at infinity, the numbers of the remaining tangents are $n - q - s$, $n - q' - s$ from the two points I, J respectively, and the number of foci is $= (n - q - s)(n - q' - s)$.

Article 104. In the case of a real curve the two points I, J are related in the same manner to the curve, and we have therefore $q = q'$; the singular tangents (if any) from the two points respectively being the same as well in character as in number. Writing $n - q - s = n - q' - s = p$, and not for the present attending to the case of singular tangents, I shall assume that the number of tangents to the curve from each of the two points is $= p$; the number of foci is thus $= p^2$; and to each focus there corresponds a directrix, viz., this is the line through the points of contact of the tangents from the focus to the two points I, J respectively.

Article 105. Consider any two foci A, B not in line with either of the points I, J , then joining these with the points I, J , and taking A_1, B_1 the intersections of AI, BJ and of AJ, BI (A_1, B_1 being therefore by a foregoing definition the antipoints of (A, B)), then A_1, B_1 are, it is clear, foci of the curve. We may out of the p^2 foci select, and that in $1 \cdot 2 \dots p$ different ways, a system of p foci such that no two of them lie in line with either of the points I, J ; and this being so, taking the antipoints of each of the $\frac{1}{2}p(p-1)$ pairs out of the p foci, we have, inclusively of the p foci, in all $p + 2 \cdot \frac{1}{2}p(p-1)$, that is p^2 foci, the entire system of foci.

111 to 117. On the Foci of Conics.

Article 106. A conic is a curve of the class 2, and the number of foci is thus $= 4$. Taking as foci any two points A, B , the remaining two foci will be the antipoints A_1, B_1 . In order that a given point A may be a focus, the conic must touch the lines AI, AJ ; similarly, in order that a given point B may be a focus, the conic must touch the lines BI, BJ ; the equation of a conic having the given points A, B for foci contains therefore a single arbitrary parameter.

Article 107. In the case, however, of the parabola the curve touches the line infinity; there is consequently from each of the points I, J only a single tangent to the curve, and consequently only one focus: the parabola having a given point A for its focus is a conic touching the line infinity and the lines AI, AJ , or say the three sides of the triangle AIJ ; its equation contains therefore two arbitrary parameters.

Article 108. Returning to the general conic, there are certain trizomal forms of the focal equation, not of any great interest, but which may be mentioned. Using circular coordinates, and taking $(\alpha, \alpha', 1)$ and $(\beta, \beta', 1)$ for the coordinates of the given foci A, B respectively, the conic touches the lines $\xi - \alpha z = 0, \eta - \alpha' z = 0, \xi - \beta z = 0, \eta - \beta' z = 0$; the equation of a conic touching the first three lines is

$$\sqrt{l(\xi - \alpha z)} + \sqrt{m(\eta - \alpha' z)} + \sqrt{n(\xi - \beta z)} = 0,$$

where l, m, n are arbitrary, and it is easy to obtain, in order that the conic may touch the fourth line $\eta - \beta' z = 0$, the condition

$$n = -\frac{lm}{l+m} \cdot \frac{(\beta - \alpha)(\beta' - \alpha')}{(\beta - \alpha) + (\beta' - \alpha')}.$$

Article 109. In fact, n having this value, the equation gives

$$l(\xi - \alpha z) + m(\eta - \alpha' z) + 2\sqrt{lm(\xi - \alpha z)(\eta - \alpha' z)} = -\frac{lm}{l+m} \frac{(\beta - \alpha)(\beta' - \alpha')}{(\beta - \alpha) + (\beta' - \alpha')} (\eta - \beta' z + (\beta' - \alpha') z),$$

and taking over the term

$$\frac{lm}{l+m} \frac{(\beta - \alpha)(\beta' - \alpha')}{(\beta - \alpha) + (\beta' - \alpha')} (\beta' - \alpha') z = \frac{lm}{l+m} (\beta - \alpha) z,$$

this gives

$$l(\xi - \beta z) + m(\eta - \alpha' z) + 2\sqrt{lm(\xi - \alpha z)(\eta - \alpha' z)} = -\frac{lm}{l+m} \frac{(\beta - \alpha)(\beta' - \alpha')}{(\beta - \alpha) + (\beta' - \alpha')} (\eta - \beta' z).$$

It is easy to see that the equation may be written in any one of the four forms

$$\begin{aligned} \sqrt{l(\xi - \alpha z)} + \sqrt{m(\eta - \beta' z)} + \sqrt{-\frac{lm}{l+m} \frac{(\beta - \alpha)(\beta' - \alpha')}{(\beta - \alpha) + (\beta' - \alpha')} (\eta - \alpha' z)} &= 0, \\ \sqrt{l(\xi - \alpha z)} + \sqrt{m(\eta - \alpha' z)} + \sqrt{-\frac{lm}{l+m} \frac{(\beta - \alpha)(\beta' - \alpha')}{(\beta - \alpha) + (\beta' - \alpha')} (\xi - \beta z)} &= 0, \\ \sqrt{l(\xi - \beta z)} + \sqrt{m(\eta - \alpha' z)} + \sqrt{-\frac{lm}{l+m} \frac{(\beta - \alpha)(\beta' - \alpha')}{(\beta - \alpha) + (\beta' - \alpha')} (\eta - \beta' z)} &= 0, \\ \sqrt{l(\xi - \beta z)} + \sqrt{m(\eta - \beta' z)} + \sqrt{-\frac{lm}{l+m} \frac{(\beta - \alpha)(\beta' - \alpha')}{(\beta - \alpha) + (\beta' - \alpha')} (\xi - \alpha z)} &= 0, \end{aligned}$$

viz., in forms containing any three of the four radicals $\sqrt{\xi - \alpha z}$, $\sqrt{\xi - \beta z}$, $\sqrt{\eta - \alpha' z}$, $\sqrt{\eta - \beta' z}$. The conic is thus expressed as a trizomal curve, the zomals being each a line, viz., they are any three out of the four focal tangents; the order of the curve, as deduced from the general expression $2^{\nu-2}r$, is $= 2$; so that there is here no depression of order.

Article 110. But the ordinary form of the focal equation is a more interesting one; viz., $\mathfrak{A}$, $\mathfrak{B}$ being as usual the squared distances of the current point from the two given foci respectively, say

$$\mathfrak{A} = (x - az)^2 + (y - a'z)^2, \quad \mathfrak{B} = (x - \beta z)^2 + (y - \beta' z)^2,$$

then $2a$ being an arbitrary parameter, the equation is

$$2az + \sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} = 0,$$

viz., the equation is here that of a trizomal curve, the zomals being curves of the second order, that is, the zomals are ($z^2 = 0$) the line infinity twice, and the line-pairs AI , AJ and BI , BJ respectively: the general expression $2^{\nu-2}r$ gives therefore the order $= 4$; but in the present case there are two branches, viz., the branches

$$2az + \sqrt{\mathfrak{A}} - \sqrt{\mathfrak{B}} = 0, \quad 2az - \sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} = 0,$$

each ideally containing ($z = 0$) the line infinity; the curve contains therefore ($2z^2 = 0$) the line infinity twice, and omitting this factor the order is $= 2$, as it should be.

Article 111. To express the equation by means of the other two foci A_1 , B_1 , writing the equation under the form

$$2\mathfrak{A} + 2\mathfrak{B} + 2\sqrt{\mathfrak{A}\mathfrak{B}} - 4a^2z^2 = 0,$$

and then if $\mathfrak{A}_1, \mathfrak{B}_1$ are the squared distances of the current point from A_1, B_1 respectively, we have (ante, No. 65),

$$\mathfrak{A}\mathfrak{B} = \mathfrak{A}_1\mathfrak{B}_1,$$

$$\mathfrak{A} + \mathfrak{B} - 2\mathfrak{A}_1 - 2\mathfrak{B}_1 = Kz^2 = K,$$

where K is the squared distance of the foci $A, B, = 4a^2e^2$ suppose: whence putting $a^2(1 - e^2) = b^2$, the equation becomes

$$2\mathfrak{A}_1 + 2\mathfrak{B}_1 + 2\sqrt{\mathfrak{A}_1\mathfrak{B}_1} - 4b^2z^2 = 0,$$

that is

$$2bz + \sqrt{\mathfrak{A}_1} + \sqrt{\mathfrak{B}_1} = 0,$$

which is the required new form. It is hardly necessary to remark that the equation $2az + \sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} = 0$, putting therein $z = 1$, and expressing $\mathfrak{A}, \mathfrak{B}$ in rectangular coordinates measured along the axes, is the ordinary focal equation $2a = \sqrt{(x - ae)^2 + y^2} + \sqrt{(x + ae)^2 + y^2}$.

Article 112. I remark that the equation $2az + \sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} = 0$ gives rise to $4a^2z^2 + \mathfrak{A} + \mathfrak{B} + 4a^2z\sqrt{\mathfrak{A}} = 0$, but here $\mathfrak{A} - \mathfrak{B} = -4aexz$, so that the equation contains $z = 0$, and omitting this it becomes $(az - ex) + \sqrt{\mathfrak{A}} = 0$, a bizomal form, being a curve of the order $= 2$, as it should be; this is in fact the ordinary equation in regard to a focus and its directrix.

118 to 123. Theorem of the Variable Zomal as applied to a Conic.

Article 113. The equation $2kz + \sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} = 0$ is in like manner that of a conic; in fact, this would be a curve of the order $= 4$, but there are as before the two branches

$$2kz + \sqrt{\mathfrak{A}} - \sqrt{\mathfrak{B}} = 0, \quad 2kz - \sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} = 0,$$

each ideally containing ($z = 0$) the line infinity, and the order is thus reduced to be $= 2$. Each of the circles $\mathfrak{A} = 0, \mathfrak{B} = 0$ is a circle having double contact with the conic (this of course implies that the centre of the circle is on an axis of the conic). We may if we please start from the form $2az + \sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} = 0$, and then by means of the theorem of the variable zomal introduce into the equation one, two, or three such circles.

Article 114. It is in this point of view that I will consider the question, viz., adapting the formula to the case of the ellipse, and starting from the form

$$2az + \sqrt{(x - aez)^2 + y^2} + \sqrt{(x + aez)^2 + y^2} = 0,$$

the equation of the variable zomal or circle of double contact may be taken to be

$$4\frac{c}{a}\sqrt{(x - qaez)^2 + y^2} + (x - \frac{c}{a}qaez)^2 + y^2 + (x + \frac{c}{a}qaez)^2 + y^2,$$

where q is an arbitrary parameter; writing for greater simplicity $z = 1$, and reducing, the equation is

$$(x - qae)^2 + y^2 = b^2(1 - q^2).$$

Article 115. If $q < 1$, then writing $q = \sin \theta$, we obtain the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

as the envelope of the variable circle

$$(x - ae \sin \theta)^2 + y^2 = b^2 \cos^2 \theta,$$

viz., of a circle having its centre on the major axis at a distance $= ae \sin \theta$ from the centre, and its radius $= b \cos \theta$. (I notice, in passing, that this gives in practice a very convenient graphical construction of the ellipse.) It may be remarked that for $\theta = \pm \sin^{-1} e$, the circle becomes

$$(x - \pm(a \mp b))^2 + y^2 = b^2,$$

viz., this is the circle of curvature at one or other of the extremities of the major axis; as θ passes from 0 to $\pm \sin^{-1} e$ we have a series of real circles, which, by their continued intersection, generate the ellipse; as θ increases from $\pm \sin^{-1} e$ to $\pm 90^\circ$, the circles continue real, but the consecutive circles no longer intersect in any real point, and ultimately for $\theta = \pm 90^\circ$, the circles become evanescent at the two foci respectively.

Article 116. In the case $q > 1$, we have a real representation of

$$(x - qae)^2 + y^2 + b^2(q^2 - 1),$$

as the squared distance of the point (x, y) from a point $(X, 0, Z)$ out of the plane of the figure, viz., putting this

$$= (x - X)^2 + y^2 + Z^2,$$

we have

$$qae = X, \quad Z^2 = b^2(q^2 - 1),$$

whence

$$\frac{X^2}{a^2 e^2} - \frac{Z^2}{b^2} = 1;$$

or, what is the same thing,

$$\frac{X^2}{a^2} - \frac{Z^2}{b^2} = e^2,$$

that is, the locus is the focal hyperbola, viz., a hyperbola in the plane of zx , having its vertices at the foci, and its foci at the vertices of the ellipse.

Article 117. If instead of the form first considered, we start from the trizomal form

$$2bz + \sqrt{ax^2 + (y - aei z)^2} + \sqrt{ax^2 + (y + aei z)^2} = 0,$$

then we have the zomal or circle of double contact under the form

$$\frac{b^2}{a^2} + (y - qaei)^2 = a^2(1 - q^2);$$

or putting herein $q = -i \tan \phi$, this is

$$x^2 + (y - ae \tan \phi)^2 = a^2 \sec^2 \phi,$$

so that we have the ellipse as the envelope of a variable circle having its centre on the minor axis of the ellipse, distance from the centre $= ae \tan \phi$, and radius $= a \sec \phi$. This is, in fact, Gergonne's theorem, according to which the ellipse is the secondary caustic or orthogonal

trajectory of rays issuing from a point and refracted at a right line into a rarer medium. It is to be remarked that for $\tan \phi = \pm \frac{b}{a}$, the equation of the circle is

$$x^2 + (y \pm (a - b))^2 = b^2,$$

viz., this is the circle of curvature at one or other extremity of the minor axis; from $\phi = 0$ to $\phi = \pm \tan^{-1} \frac{b}{a}$, the intersections of the consecutive circles are real, and give the entire real ellipse; from $\phi = \pm \tan^{-1} \frac{b}{a}$ to $\phi = \pm 90$, the circles are still real, but the intersections of consecutive circles are imaginary.

Article 118. If in the equation of the generating circle we interchange x, y, a, b , the equation becomes

$$(x - ae \tan \phi)^2 + y^2 = b^2 \sec^2 \phi,$$

which is (as it should be) equivalent to the former equation

$$(x - ae \sin \theta)^2 + y^2 = b^2 \cos^2 \theta,$$

the identity being established by means of the equation

$$\cos \theta = \frac{\cos \phi}{\sqrt{1 - e^2 \sin^2 \phi}}, \text{ and therefore } \sin \theta = i \tan \phi \sqrt{1 - \frac{\cos^2 \phi}{1 - e^2 \sin^2 \phi}}, \tan \theta = i \sin \phi \sqrt{1 - e^2 \sin^2 \phi}$$

which is Jacobi's imaginary transformation in the theory of Elliptic Functions.

124 to 126. Foci of the Circular Cubic and the Bicircular Quartic.

Article 119. For a cubic curve, the class is in general = 6, and the number of the foci is = 36. But a specially interesting case is that of a circular cubic, viz., a cubic passing through each of the circular points at infinity. Here, at each of the circular points at infinity, the tangent at this point reckons twice among the tangents to the curve from the point; the number of the remaining tangents is thus = 4, and the number of the foci is = 16. If from any two points whatever on the curve tangents be drawn to the curve, then the two pencils of tangents are, and that in four different ways, homologous to each other, viz., if the tangents of the first pencil are (1, 2, 3, 4), and those of the second pencil, taken in a proper order, are (1', 2', 3', 4'), then we have (1, 2, 3, 4) homologous with each of the arrangements (1', 2', 3', 4'), (2', 1', 4', 3'), (3', 4', 1', 2'), (4', 3', 2', 1'). And in each case the intersections of the four corresponding tangents lie on a conic passing through the two given points on the curve.

Article 120. Hence taking the points on the curve to be the circular points at infinity, we have the sixteen foci lying in fours upon four different circles—that is, we have four tetrads of concyclic foci. Let any one of these tetrads be A, B, C, D , then if

$$\begin{aligned} \text{Antipoints of } (B, C), (A, D) &\text{ are } (B_1, C_1), (A_1, D_1), \\ (C, A), (B, D) &\text{ are } (C_2, A_2), (B_2, D_2), \\ (A, B), (C, D) &\text{ are } (A_3, B_3), (C_3, D_3), \end{aligned}$$

the four tetrads of concyclic foci are

$$\begin{array}{cccc} A, & B, & C, & D; \\ A_1, & B_1, & C_1, & D_1; \\ A_2, & B_2, & C_2, & D_2; \\ A_3, & B_3, & C_3, & D_3. \end{array}$$

It is to be observed that if A, B, C, D are any four points on a circle, then if, as above, we pair these in any manner, and take the antipoints of each pair, the four antipoints lie on a circle, and thus the original system A, B, C, D , of four points on a circle, leads to the remaining three systems of four points on a circle. The theory is in fact that already discussed ante, No. 72 *et seq.*

Article 121. The preceding theory applies without alteration to the bicircular quartic, viz., the quartic curve which has a node at each of the circular points at infinity. The class is here = 8, but among the tangents from a node each of the two tangents at the node is to be reckoned twice, and the number of the remaining tangents is = 4: the number of foci is = 16. And, by the general theorem that in a binodal quartic the pencils of tangents from the two nodes respectively are homologous, the sixteen foci are related to each other precisely in the manner of the foci of the circular cubic. The latter is in fact a particular case of the former, viz., the bicircular quartic may break up into the line infinity, and a circular cubic.

127 to 129. Centre of the Circular Cubic, and Nodo-Foci, &c. of the Bicircular Quartic.

Article 122. The tangents at I, J have not been recognised as tangents from I, J , giving by their intersection a focus, but it is necessary in the theory to pay attention to the tangents in question. It is clear that these tangents are in fact asymptotes; viz., in the case of the circular cubic they are the two imaginary asymptotes of the curve, and in the case of a bicircular quartic, the two pairs of imaginary parallel asymptotes; but it is convenient to speak of them as the tangents at I, J .

Article 123. In the case of a circular cubic, the tangents at I and J meet in a point which I call the *centre* of the curve, viz., this is the intersection of the two imaginary asymptotes. In the case of a bicircular quartic, the tangents at I consist of a pair of lines meeting in a point which I call a *nodo-focus*, and the tangents at J of a pair of lines meeting in another point which I call the other nodo-focus; the two nodo-foci are antipoints of each other, and the line joining them passes through the centre of the curve.

Article 124. The nodo-foci of a bicircular quartic are to be distinguished from the ordinary foci; the nodo-foci are the intersections of the tangents at the circular points at infinity, while the ordinary foci are the intersections of the tangents from the circular points at infinity to the curve. The nodo-foci are, however, included among the foci in the wider sense, viz., they are the points which correspond to the case where the tangents from I and J are the tangents at these points respectively.

130 to 157. Additional Investigations on the Theory of Foci.

Article 125. The theory of foci may be further developed by considering the properties of the polyzomal curves in relation to the circular points at infinity. I have already shown

that the general trizomal curve $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$ may be expressed by means of variable zomals, and that when the zomals are circles, the curve belongs to the theory of foci. I proceed to consider some further developments of this theory.

Article 126. If the zomals of a trizomal curve are circles, then the curve is in general a bicircular quartic; for the order of the trizomal is $= 2r$, and if $r = 2$, the order is $= 4$; and since each zomal is a circle, the curve passes through the circular points at infinity, and has a node at each of these points. The trizomal curve $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$, where $U = 0$, $V = 0$, $W = 0$ are circles, is thus a bicircular quartic.

Article 127. The foci of a bicircular quartic may be determined as follows: The tangents from the circular points at infinity to the curve are the lines joining these points to the foci; and since the curve is a trizomal with circular zomals, the tangents from I and J may be determined by means of the equations of the zomals. In particular, if the zomals are circles passing through two fixed points, then the foci are the antipoints of these fixed points.

Article 128. The case where the zomals are circles having a common radical axis is particularly interesting; for in this case the trizomal curve degenerates into a pair of circles, or into a circular cubic and the line infinity. The condition for this degeneration is that the Jacobian of the three circles passes through the circular points at infinity.

Article 129. I remark that the focal equation of a conic may be generalised to the case of a bicircular quartic. If $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ are the squared distances of the current point from four foci, then the equation

$$a\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}} + d\sqrt{\mathfrak{D}} = 0,$$

where $\frac{a}{l} + \frac{b}{m} + \frac{c}{n} + \frac{d}{p} = 0$, represents a bicircular quartic having the four given points for foci. This is the generalisation of the focal equation of a conic to the case of a quartic curve.

Article 130. The theory of polyzomal curves is thus seen to be intimately connected with the theory of foci, and the results obtained in the present memoir furnish a new method of investigating the properties of curves defined by their foci. The method is particularly applicable to the circular cubic and the bicircular quartic, and it is probable that further developments of the theory will lead to important results in regard to these and other curves.

Article 131. I conclude this part of the memoir by remarking that the theory of the variable zomal, which has been established for the trizomal curve, may be extended to the general ν -zomal curve. Given the ν -zomal curve $\sqrt{U_1} + \sqrt{U_2} + \dots + \sqrt{U_\nu} = 0$, if the zomals are connected by an identical relation $a_1U_1 + a_2U_2 + \dots + a_\nu U_\nu = 0$, then the curve may be expressed by means of any $\nu - 1$ of the zomals, and the equation will contain a variable parameter. This is the general theorem of the variable zomal, of which the theorem for the trizomal is a particular case.

Article 132. In the case of the tetrazomal curve, the theorem of the variable zomal does not in general hold; for, as has been shown, the tetrazomal curve does not in general admit of being expressed by means of a new zomal in addition to three of the original zomals. There are, however, particular cases in which the theorem does hold, and these cases are of considerable interest in the theory of curves.

Article 133. The present memoir contains the foundations of a new theory of curves, which I have called the theory of polyzomal curves. The theory is far from being complete, and

there are many questions which remain to be investigated. I hope, however, that the results which have been obtained will be found to be of sufficient interest to justify the publication of this preliminary account of the theory.